



NONLINEAR PHYSICS AND MECHANICS

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Geometry and Kinematics of the Mecanum Wheel on a Plane and a Sphere

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This article is devoted to a study of the geometry and kinematics of the Mecanum wheels, also known as Ilon wheels or the Swedish wheels. The Mecanum wheels are one of the types of omnidirectional wheels. This property is provided by peripheral rollers whose axes are deviated from the wheel one by 45 degrees. A unified approach to studying the geometry and kinematics of the Mecanum wheels on a plane and on the internal or external surface of a sphere is proposed. Kinematic relations for velocities at the contact point of the wheel and the supporting surface, and angular velocities of the roller relative to the supporting surface are derived. They are necessary to describe the dynamics of the Mecanum systems taking into account forces and moments of contact friction in the presence of slipping. From the continuous contact condition, relations determining the geometry of the wheel rollers on a plane and on the internal or external surface of a sphere are obtained. The geometric relations for the Mecanum wheel rollers could help to adjust the existing shape of the Mecanum wheel rollers of spherical robots and ballbots to improve the conditions of contact between the rollers and the spherical surface. An analytical study of the roller geometry was carried out, and equations of their generatrices were derived. Under the no-slipping condition, expressions for rotational velocities of the wheel and the contacting roller are obtained. They are necessary for analyzing the motion of systems within the framework of nonholonomic models, solving problems of controlling Mecanum systems and improving its accuracy. Using the example of a spherical robot with an internal three-wheeled Mecanum platform, the influence of the rollers on the robot movement was studied at the kinematic level. It has been established that the accuracy of the robot movement is influenced not only by the geometric parameters of the wheels and the number of rollers, but also by the relationship between the components of the platform center velocity and its angular velocity. Results of the numerical simulation of the motion of the spherical robot show a decrease in control accuracy in

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the absence of feedback on the robot's position due to effects associated with the finite number of rollers, their geometry and switching. These effects lead not only to high-frequency vibrations, but also to a “drift” of the robot trajectory relative to the reference trajectory. Further research on this topic involves the use of the motion separation methods and the statistical methods for kinematical and dynamical analysis of Mecanum systems.

Keywords: Mecanum wheel, omnidirectional wheel, omniwheel, spherical robot, ballbot, forward kinematics, inverse kinematics, kinematic model

1. Introduction

Mecanum wheels, invented in 1972 by the Swedish engineer B. E. Ilon [1], are one of the types of omnidirectional motion wheels [2]. On the periphery of each of them there are rollers, the axes of which intersect with the wheel axis at an angle of 45° (for the “classic” universal omniwheels this angle is 90°).

The design of Mecanum wheels allows a vehicle equipped with them to move in any direction with any orientation, make a turn on the spot, etc. However, when vehicles with the Mecanum wheels move, vibrations arise [3, 4]. The source of these is the switching of the contacting rollers and the imperfection of their geometric shape.

Due to its mobility and ease of parking, omnidirectional platforms are used for working in cramped conditions of warehouses, production facilities and similar premises, and for creating vehicles for people with disabilities and for other purposes [3, 5].

Omnidirectional wheels of various types are used to control the motion of spherical robots, and spherical wheels and bodies. The motion of such systems has been studied in [6–12]. For spherical robots and vehicles with a spherical wheel, it is necessary to correct the shape of the Mecanum wheel rollers to improve the quality of the contact [9, 11]. Examples of the Mecanum systems are shown in Fig. 1.

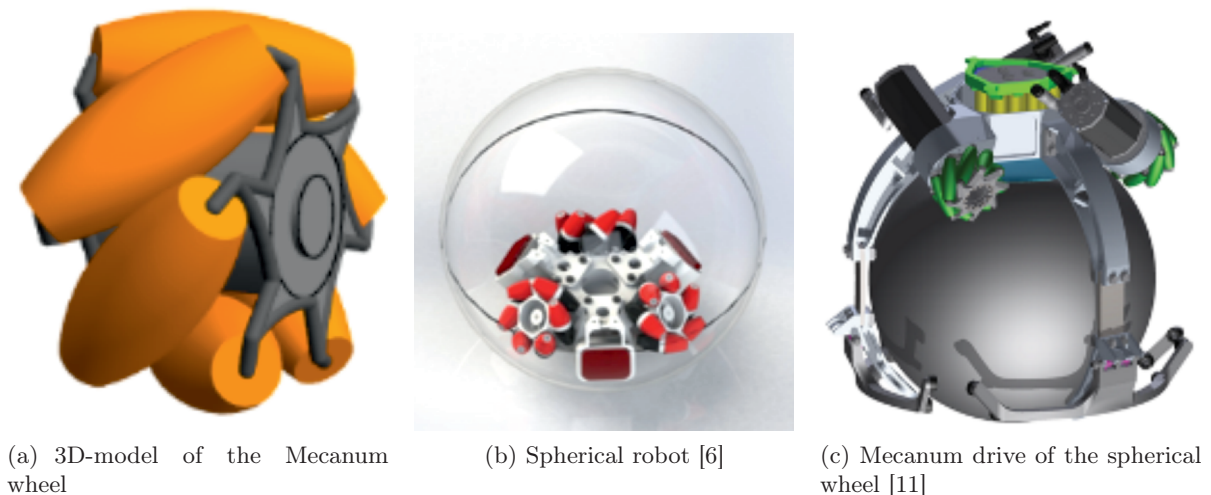


Fig. 1. Examples of the systems with the Mecanum wheels

In most cases, an omni- or Mecanum wheel is modeled as a disk sliding in a direction perpendicular to the axis of the contact roller. According to [2], the first studies of the Mecanum

wheels using this model were published by J. Agullo et al. [13], P. Muir and C. P. Neuman [14] in 1987. The equations of kinematics and dynamics of vehicles with an arbitrary number and location of the omni- or Mecanum wheels on a plane or a sphere were derived by A. V. Borisov et al. [15]. The application of V. Y. Tatarinov's equations to study the dynamics of an omnivehicle with an arbitrary arrangement and type of wheels is described in [16] by A. A. Zobova.

The sliding disk model is convenient for kinematic analysis and synthesis. However, when describing the dynamics of a Mecanum system, the use of this model does not allow us to take into account many effects associated with the movement and switching of rollers.

If we assume that the rollers are always in contact with the supporting surface with a central cross section [17, 18], then it is possible to take into account, for example, friction on their axes.

The first comprehensive study of the geometry, kinematics and dynamics of Mecanum systems taking into account rollers was published by G. Wampfler et al. [19] in 1989. The relations for the coordinates of the roller contact point along the wheel axis, and the relations for the velocities at the contact point were obtained. At the stage of expressing the speeds of rotation of wheels and rollers from the condition of no slippage, an unfortunate mistake was made — in some parts of the expressions, $\sin \alpha$ and $\cos \alpha$ were confused (possibly due to a change in the method of calculating the angle α of deviation of the roller axis from the wheel one).

A comprehensive study of the geometry of the rollers and the kinematics of the Mecanum wheels was carried out by A. Gferrer [20] in 2010. Using descriptive geometry methods, an equation to determine the position of the contact point and a formula for the generatrix of the roller surface, expressed in terms of the angle of rotation of the wheel, were obtained. The kinematics of the Mecanum wheels was studied taking into account the motion of the rollers.

An analysis of the influence of the design of Mecanum wheels on the accuracy of odometric navigation of the mobile robot was carried out in [21]. An improved navigation algorithm is proposed by averaging the kinematic equations over the angle of rotation of the wheel.

A study of the influence of the Mecanum wheel design on the dynamics of the omnipatform taking into account the slippage and the multicomponent contact friction was carried out in [22].

An approximation of the rollers by a set of spheres is proposed in [23]. The resulting model with two-point contact between the Mecanum wheels and the floor describes the motion of the four-wheel robot more accurately than the single-point contact model.

The “intuitive” generatrix of the Mecanum wheel roller surface is the arc of an ellipse or a circle [24–26].

With an exact (ideal) roller shape, the projection of the Mecanum wheel is a perfect circle, i. e., it can roll on a plane without vertical movements of its center.

A. Gferrer [20] showed that the exact generatrix of the roller is not elliptical and also proposed a method to approximate the roller with a torus. Descriptions of the Mecanum wheel roller geometry equivalent to [20] are given in [27, 28].

A study of the dependence of the magnitude of vertical vibrations on the curvature radius of the Mecanum wheel roller generatrices is given in [26]. Proposals are given for modifying the wheel design by adding the elastic elements to dampen vibrations.

In [28], an equation of the Mecanum wheel roller moving on a plane was derived. The conditions of tangency of the projections of the roller cross sections (ellipses) to the projection of the wheel (circle) on the plane xz , orthogonal to the axis of the wheel y , were used. In the case of a Mecanum wheel rolling on the external surface of the ball, the resulting equations were modified by adding the correction terms. In this case, the geometry analysis is approximate (see Appendix A), no rigorous study of the conditions of contact between the roller and the sphere

was carried out. The resulting shape of the rollers was used in the design of the Mecanum drive of the spherical wheel of the vehicle [11] (see Fig. 1c).

It should be noted that the geometry of the rollers of the Mecanum wheels, designed to move on a flat surface, has been studied in sufficient detail. The problem of synthesizing the shape of the Mecanum wheel rollers on a spherical surface requires further solutions.

The results of the study of the dynamics and modeling of the motion of the vehicles with the “classic” omniwheels, taking into account the nature of the contact of the rollers with the floor, are presented in [29–31].

In [29], the authors put forward the curious thesis that a real omniwheel is an approximate implementation of a theoretical one (i. e., a sliding disk), and not vice versa.

The aims of this study are:

- to determine the shape of the rollers of the Mecanum wheels for moving on a plane or a sphere within a unified approach;
- to obtain kinematic relationships for the wheel rotational speeds, necessary to solve control problems for Mecanum systems moving on a plane or a sphere;
- to obtain relationships for the velocities at the point of contact of the wheels with the supporting surface, and the angular velocities of the rollers, necessary for studying the dynamics of Mecanum systems, taking into account wheel slipping, the structure of the contact friction forces and moments.

REMARK 1. Further, if necessary, we use the notation $\mathfrak{c}(\cdot) = \cos(\cdot)$, $\mathfrak{s}(\cdot) = \sin(\cdot)$.

2. Description of the Mecanum wheel. Motion on a plane

The Mecanum wheel under consideration consists of a hub and N rollers, evenly distributed along its periphery. The rollers passively rotate about their axes, intersecting with the wheel axis at a constant angle (for example, 45° for a Mecanum wheel and 90° for a universal omniwheel).

Let us introduce a coordinate system $CXYZ$ with origin at the center of the wheel C (Fig. 2). CY -axis is an axis of rotation of the wheel, CXZ is a middle plane of the wheel. In all cases under consideration, the movement occurs in such a way that the Z -axis is always normal to the wheel supporting surface (a plane or a sphere, Fig. 2).

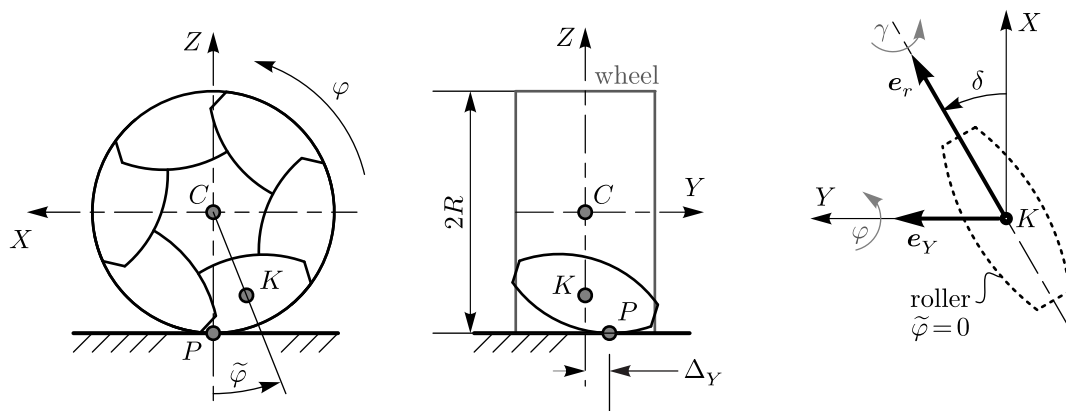


Fig. 2. Scheme of the wheel on a plane

Let φ be the angle of rotation of the wheel about its own CY -axis, and γ be the angle of rotation of the roller about its own axis (with a unit vector $\mathbf{e}_r$, Fig. 2).

To facilitate further description, we introduce the angle of rotation of the contacting roller, $\tilde{\varphi}$, about the wheel CY -axis (Fig. 2):

$$\tilde{\varphi} = \angle(-Z, \overrightarrow{CK}), \quad -\frac{\pi}{N} \leq \tilde{\varphi} < \frac{\pi}{N},$$

where K is the contacting roller center. As the contacting roller is changed, the value of $\tilde{\varphi}$ jumps from $\frac{\pi}{N}$ to $-\frac{\pi}{N}$ (or vice versa, depending on the rotation direction).

The expression of $\tilde{\varphi}$ in terms of the wheel rotation angle φ can be determined using mod operation of taking the remainder of division according to the formula

$$\tilde{\varphi} = \left(\left(\varphi + \frac{\pi}{N} \right) \bmod \frac{2\pi}{N} \right) - \frac{\pi}{N}.$$

The deviation angle of the contacting roller axis from the plane CXZ

$$\delta = \angle(XZ, \mathbf{e}_r) = \angle(X, \mathbf{e}_r)|_{\tilde{\varphi}=0}.$$

For a Mecanum wheel it is usually $|\delta| = 45^\circ$, and for a universal omniwheel, $\delta = 0$.

The set of rollers of an ideal Mecanum wheel for motion on a plane is enveloped by a cylindrical surface of radius R with CY -axis (Fig. 3). This enables the wheel to roll and spin, maintaining a constant distance from the center C to the supporting surface and a constant orientation of the CZ -axis.

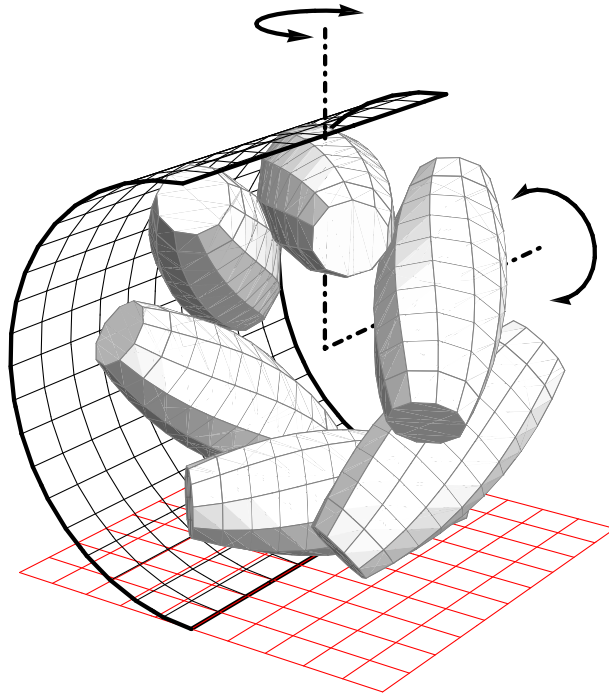


Fig. 3. Cylindrical envelope of the Mecanum wheel rollers on a plane (the roller “meridians” are the lines of contact between the roller surface and the envelope surface; the possible directions and axes of the wheel rotation relative to the plane are shown)

REMARK 2. Here and in what follows it is assumed that the surfaces of all the rollers simultaneously touch the envelope surface at any angle of rotation of the wheel.

It also follows from this that during the movement of the wheel, the contact point P of the roller with the support surface is displaced relative to the wheel center C only in the direction of the CY -axis, but not the axis CX . The axial displacement of point P is denoted by Δ_Y in Fig. 2.

Let us find the velocity of the roller contact point P :

$$\mathbf{V}_P = \mathbf{V}_C + \boldsymbol{\Omega}_{cf} \times \overrightarrow{CP} + \dot{\varphi} \mathbf{e}_Y \times \overrightarrow{CP} + \dot{\gamma} \mathbf{e}_r \times \overrightarrow{KP}, \quad (2.1)$$

where $\mathbf{V}_C = (V_{C_X} \ V_{C_Y} \ 0)^\top$ is the velocity of the wheel center C ; $\boldsymbol{\Omega}_{cf} = (0 \ 0 \ \Omega_Z)^\top$ is the angular velocity of the XYZ -frame, Ω_Z is the velocity of the wheel spinning; $\overrightarrow{CP} = (0 \ \Delta_Y \ -R)^\top$; $\mathbf{e}_Y = (0 \ 1 \ 0)^\top$ is the unit vector of the wheel axis; and $\mathbf{e}_r$ is the unit vector of the roller axis. Here all vectors are given in the XYZ -frame.

We find the coordinates of the vectors $\overrightarrow{CK}$, $\mathbf{e}_r$ by multiplying the rotation matrix by the angle $\tilde{\varphi}$ by the coordinate columns of the corresponding vectors at $\tilde{\varphi} = 0$:

$$\overrightarrow{CK} = \begin{bmatrix} \cos \tilde{\varphi} & 0 & \sin \tilde{\varphi} \\ 0 & 1 & 0 \\ -\sin \tilde{\varphi} & 0 & \cos \tilde{\varphi} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ -R_H \end{pmatrix} = \begin{pmatrix} -R_H \sin \tilde{\varphi} \\ 0 \\ -R_H \cos \tilde{\varphi} \end{pmatrix}; \quad (2.2)$$

$$\mathbf{e}_r = \begin{bmatrix} \cos \tilde{\varphi} & 0 & \sin \tilde{\varphi} \\ 0 & 1 & 0 \\ -\sin \tilde{\varphi} & 0 & \cos \tilde{\varphi} \end{bmatrix} \begin{pmatrix} \cos \delta \\ \sin \delta \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \tilde{\varphi} \cos \delta \\ \sin \delta \\ -\sin \tilde{\varphi} \cos \delta \end{pmatrix}. \quad (2.3)$$

Here $R_H = CK$ is the wheel hub “radius”.

The vector $\overrightarrow{KP}$ is

$$\overrightarrow{KP} = \overrightarrow{CP} - \overrightarrow{CK} = \begin{pmatrix} R_H \sin \tilde{\varphi} \\ \Delta_Y \\ R_H \cos \tilde{\varphi} - R \end{pmatrix}.$$

Having performed the necessary transformations in (2.1), we find the components of the velocity of the contact point P :

$$V_{P_X} = V_{C_X} - \Delta_Y \Omega_Z - R \dot{\varphi} + \dot{\gamma} \Delta_Y \sin \tilde{\varphi} \cos \delta + \dot{\gamma} (R_H \cos \tilde{\varphi} - R) \sin \delta, \quad (2.4)$$

$$V_{P_Y} = V_{C_Y} + \dot{\gamma} (R \cos \tilde{\varphi} - R_H) \cos \delta, \quad (2.5)$$

$$V_{P_Z} = \dot{\gamma} (\Delta_Y \cos \tilde{\varphi} \cos \delta - R_H \sin \tilde{\varphi} \sin \delta). \quad (2.6)$$

Since $V_{P_Z} \equiv 0$, from the last relation we obtain the axial displacement of the contact point

$$\Delta_Y = R_H \tan \delta \tan \tilde{\varphi}. \quad (2.7)$$

This equation is equivalent to the results [19, 20].

Substituting expression (2.7) into (2.4) and (2.5) gives the following formulas for the velocity components at the contact point P :

$$V_{P_X} = V_{C_X} - \Omega_Z R_H \tan \tilde{\varphi} \tan \delta - R\dot{\varphi} - \dot{\gamma} \left(R - \frac{R_H}{\cos \tilde{\varphi}} \right) \sin \delta, \quad (2.8)$$

$$V_{P_Y} = V_{C_Y} + \dot{\gamma} (R \cos \tilde{\varphi} - R_H) \cos \delta. \quad (2.9)$$

Let us write the last relations in a simplified form, expanding in $\tilde{\varphi}$ up to the second degree inclusive:

$$V_{P_X} = V_{C_X} - \Omega_Z R_H \tilde{\varphi} \tan \delta - R\dot{\varphi} - \dot{\gamma} \left(R - R_H - \frac{1}{2} R_H \tilde{\varphi}^2 \right) \sin \delta, \quad (2.10)$$

$$V_{P_Y} = V_{C_Y} + \dot{\gamma} \left(R - R_H - \frac{1}{2} R_H \tilde{\varphi}^2 \right) \cos \delta. \quad (2.11)$$

Express the rotational speeds of the wheel $\dot{\varphi}$ and the contact roller $\dot{\gamma}$ in the absence of slippage. The conditions $V_{P_X} = 0$, $V_{P_Y} = 0$ for (2.8) and (2.9) are achieved when

$$\dot{\varphi} = \frac{1}{R} \left(V_{C_X} + \frac{V_{C_Y} \tan \delta}{\cos \tilde{\varphi}} - R_H \Omega_Z \tan \delta \tan \tilde{\varphi} \right), \quad (2.12)$$

$$\dot{\gamma} = - \frac{V_{C_Y}}{(R \cos \tilde{\varphi} - R_H) \cos \delta}. \quad (2.13)$$

Let us write the last relations in a simplified form, expanding in $\tilde{\varphi}$ up to the second degree inclusive:

$$\dot{\varphi} = \frac{1}{R} \left[V_{C_X} + V_{C_Y} \left(1 + \frac{\tilde{\varphi}^2}{2} \right) \tan \delta - \Omega_Z R_H \tilde{\varphi} \tan \delta \right], \quad (2.14)$$

$$\dot{\gamma} = - \frac{V_{C_Y}}{(R - R_H) \cos \delta} \left(1 + \frac{R}{R - R_H} \frac{\tilde{\varphi}^2}{2} \right). \quad (2.15)$$

Note that from Eq. (2.13) or (2.15) it follows that the roller rotates only if there is an axial component of the wheel center velocity V_{C_Y} . When $V_{C_Y} = 0$, the Mecanum wheel or the omniwheel operates like a conventional one.

REMARK 3. In the absence of a displacement of the roller contact point at $\tilde{\varphi} = 0$, the formula for $\dot{\varphi}$ takes the form obtained for the simplest model of a Mecanum wheel — a disk sliding in the direction perpendicular to the contacting roller axis [13, 15]:

$$R\dot{\varphi} = V_{C_X} + V_{C_Y} \tan \delta.$$

We complete this section by expressing the components of the contacting roller's angular velocity $\boldsymbol{\Omega}_r = \boldsymbol{\Omega}_{cf} + \dot{\varphi} \mathbf{e}_Y + \dot{\gamma} \mathbf{e}_r$:

$$\Omega_{r_X} = \dot{\gamma} \cos \delta \cos \tilde{\varphi}, \quad \Omega_{r_Y} = \dot{\varphi} + \dot{\gamma} \sin \delta, \quad \Omega_{r_Z} = \Omega_Z - \dot{\gamma} \cos \delta \sin \tilde{\varphi}, \quad (2.16)$$

and also give their expansions in $\tilde{\varphi}$:

$$\Omega_{r_X} = \dot{\gamma} \left(1 - \frac{\tilde{\varphi}^2}{2} \right) \cos \delta, \quad \Omega_{r_Y} = \dot{\varphi} + \dot{\gamma} \sin \delta, \quad \Omega_{r_Z} = \Omega_Z - \dot{\gamma} \tilde{\varphi} \cos \delta.$$

3. Kinematics of the Mecanum wheel on the internal surface of a sphere

We now turn our attention to the Mecanum wheel motion on the internal surface of a spherical shell. In this case, the geometry of the rollers differs from that discussed earlier.

The set of rollers of an ideal Mecanum wheel for moving on the internal surface of a sphere is enveloped by a self-intersecting torus surface formed by rotation of an arc of the sphere about the CY -axis of the wheel (Fig. 4). This enables the wheel to roll and spin, maintaining a constant distance from the center C to the supporting surface and a constant orientation of the CZ -axis, and during motion the contact point of the roller P is displaced relative to C only in the direction of the CY -axis. To describe the displacement, we introduce the angle $\alpha = \angle(\vec{OC}, \vec{OP})$ (see Fig. 5).

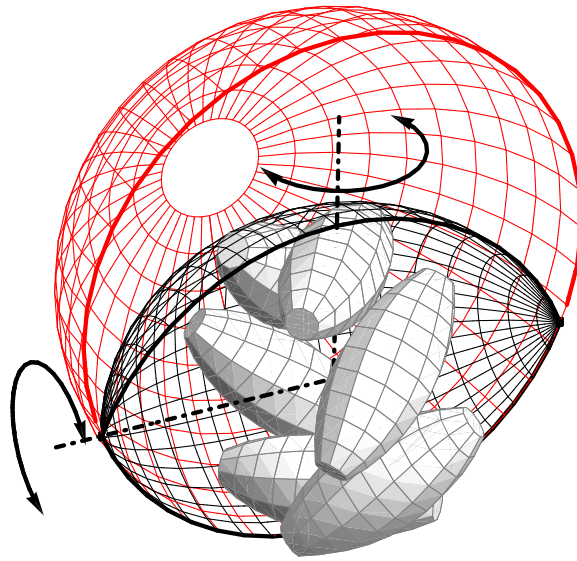


Fig. 4. Toric envelope of the Mecanum wheel rollers on the internal surface of a sphere (the roller “meridians” are the lines of contact between the roller surface and the envelope surface; the possible directions and axes of the wheel rotation relative to the sphere are shown)

Let R_S denote the radius of the spherical shell inside which the wheel moves (Fig. 5), and let $R_T = OC = \text{const}$ be the distance from the center of the wheel to the center of the sphere. Let us call their difference

$$R = R_S - R_T$$

the *radius of the wheel*.

Introduce the radius ratios

$$\lambda = \frac{R_S}{R_T}, \quad \kappa = \frac{R_H}{R_T},$$

where $R_H = CK$ is the hub “radius”. Here $\lambda > 1$, $\kappa > 0$.

Consider the kinematics of the wheel motion relative to the sphere.

The velocity at the contact point P relative to the spherical shell has the form

$$\mathbf{V}_P = \mathbf{V}_C + \boldsymbol{\Omega}_{cf} \times \vec{CP} + \dot{\varphi} \mathbf{e}_Y \times \vec{CP} + \dot{\gamma} \mathbf{e}_r \times \vec{KP}, \quad (3.1)$$

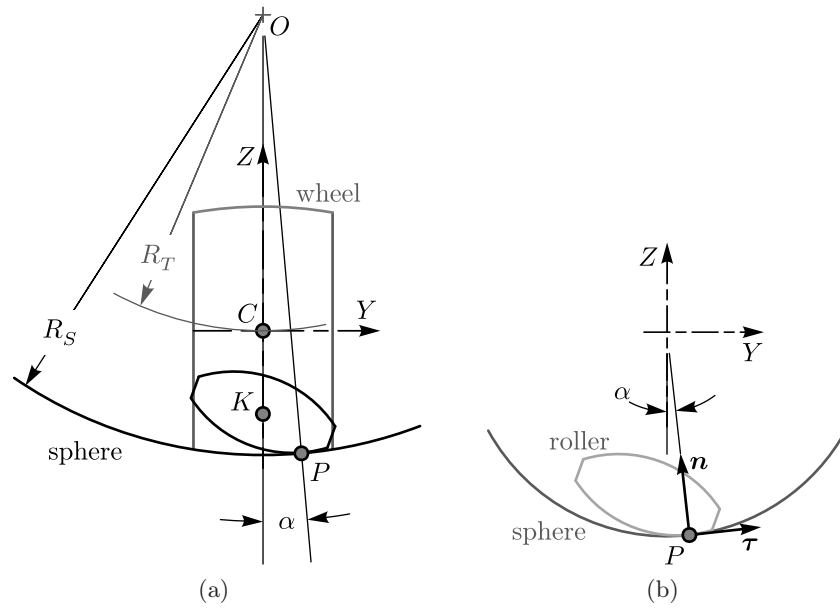


Fig. 5. Scheme of the internal wheel in a sphere

where $\mathbf{V}_C = (V_{C_x} \ V_{C_y} \ 0)^\top$ is the relative velocity of the center C ; $\boldsymbol{\Omega}_{cf} = \left(\frac{V_{C_y}}{R_T} \ -\frac{V_{C_x}}{R_T} \ \Omega_Z \right)^\top$ is the angular velocity of the XYZ -frame relative the sphere;

$$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \begin{pmatrix} 0 \\ R_S \sin \alpha \\ -R_S \cos \alpha \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ -R_T \end{pmatrix} = \begin{pmatrix} 0 \\ R_S \sin \alpha \\ -R_S \cos \alpha + R_T \end{pmatrix}; \quad (3.2)$$

$$\overrightarrow{KP} = \overrightarrow{CP} - \overrightarrow{CK} = \begin{pmatrix} R_H \sin \tilde{\varphi} \\ R_S \sin \alpha \\ R_H \cos \tilde{\varphi} - R_S \cos \alpha + R_T \end{pmatrix}; \quad (3.3)$$

the vectors $\overrightarrow{CK}$ and $\mathbf{e}_r$ are defined by formulas (2.2) and (2.3).

Having performed the necessary transformations in (3.1), we find the components of the velocity of the contact point P :

$$\begin{aligned} V_{P_x} &= \frac{R_S}{R_T} V_{C_x} c\alpha - R_S \Omega_Z s\alpha + \dot{\varphi}(R_T - R_S c\alpha) - \\ &\quad - \dot{\gamma}((-R_T - R_H c\tilde{\varphi} + R_S c\alpha) s\delta - R_S s\alpha c\delta s\tilde{\varphi}), \\ V_{P_y} &= \frac{R_S}{R_T} V_{C_y} c\alpha + \dot{\gamma}(-(R_T - R_S c\alpha) c\tilde{\varphi} - R_H) c\delta, \\ V_{P_z} &= \frac{R_S}{R_T} V_{C_y} s\alpha - \dot{\gamma}(R_H s\delta s\tilde{\varphi} - R_S s\alpha c\delta c\tilde{\varphi}). \end{aligned} \quad (3.4)$$

To analyze the conditions of contact between the rollers and the sphere, we find the projections of the relative velocities onto the normal to the sphere $\mathbf{n} = (0 \ -\sin \alpha \ \cos \alpha)^\top$ and onto the tangents X , $\boldsymbol{\tau} = (0 \ \cos \alpha \ \sin \alpha)^\top$ (see Fig. 5).

Project the velocity $\mathbf{V}_P$ onto the tangent and the normal to the sphere. The tangent component in the YZ -plane is

$$\begin{aligned} V_{P\tau} &= \mathbf{V}_P \cdot \boldsymbol{\tau} = V_{P_Y} c\alpha + V_{P_Z} s\alpha = \\ &= \frac{R_S}{R_T} V_{C_Y} - \dot{\gamma}((R_H + R_T c\tilde{\varphi}) c\alpha c\delta + R_H s\alpha s\delta s\tilde{\varphi} - R_S c\delta c\tilde{\varphi}). \end{aligned} \quad (3.5)$$

The normal component of the contact point velocity is

$$V_{Pn} = \mathbf{V}_P \cdot \mathbf{n} = -V_{P_Y} s\alpha + V_{P_Z} c\alpha = -\dot{\gamma}((-R_T c\tilde{\varphi} - R_H) s\alpha c\delta + R_H s\tilde{\varphi} c\alpha s\delta).$$

From the condition of continuous contact between the wheel and the sphere at $V_{Pn} = 0$ we obtain

$$\begin{aligned} \tan \alpha &= \frac{R_H \tan \delta \sin \tilde{\varphi}}{R_T \cos \tilde{\varphi} + R_H}, \\ R_T \tan \alpha &= \frac{R_H \tan \delta \tan \tilde{\varphi}}{1 + \kappa \sec \tilde{\varphi}}. \end{aligned} \quad (3.6)$$

Note that, as $R_S \rightarrow \infty$, the sphere degenerates into a plane, $\kappa = \frac{R_H}{R_T} \rightarrow 0$, and $R_T \tan \alpha \rightarrow \Delta_Y$. The shape of the wheel rollers in the limiting case corresponds to the results obtained above: for $R_S \rightarrow \infty$ Eq. (3.6) becomes (2.7):

$$\Delta_Y = R_T \tan \alpha = R_H \tan \delta \tan \tilde{\varphi}.$$

Substituting into the kinematic relations, we find

$$\cos \alpha = \frac{1}{\sqrt{1 + \tan^2 \alpha}} = \frac{1 + \kappa \cos \tilde{\varphi}}{\sqrt{(1 + \kappa \cos \tilde{\varphi})^2 + (\kappa \tan \delta \sin \tilde{\varphi})^2}}, \quad (3.7)$$

$$\sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}} = \frac{\kappa \tan \delta \sin \tilde{\varphi}}{\sqrt{(1 + \kappa \cos \tilde{\varphi})^2 + (\kappa \tan \delta \sin \tilde{\varphi})^2}}. \quad (3.8)$$

Approximately,

$$\cos \alpha \approx 1 - \frac{\tilde{\varphi}^2 \kappa^2 \tan^2 \delta}{2(1 + \kappa)^2}, \quad \sin \alpha \approx \frac{\tilde{\varphi} \kappa \tan \delta}{1 + \kappa}.$$

Substitution of expressions (3.7) and (3.8) for the angle α in Eqs. (3.4) and (3.5) for V_{P_X} , $V_{P\tau}$ gives cumbersome results — we do not present them here. We limit ourselves only to approximations in the angle $\tilde{\varphi}$ with an accuracy of $O(\tilde{\varphi}^3)$

$$\begin{aligned} V_{P_X} &= \lambda \left(1 - \frac{\kappa^2}{(1 + \kappa)^2} \frac{\tilde{\varphi}^2}{2} \tan^2 \delta \right) V_{C_X} - \\ &- \frac{\lambda}{1 + \kappa} R_H \Omega_Z \tilde{\varphi} \tan \delta - \dot{\varphi} \left(R - \frac{\lambda \kappa}{(1 + \kappa)^2} \frac{R_H \varphi^2}{2} \tan^2 \delta \right) - \end{aligned} \quad (3.9)$$

$$\begin{aligned} &- \dot{\gamma} \left(R - R_H - \left(\frac{2\lambda}{1 + \kappa} + \frac{\lambda \kappa}{(1 + \kappa)^2} \tan^2 \delta - 1 \right) \frac{R_H \varphi^2}{2} \right) \sin \delta, \\ V_{P\tau} &= \lambda V_{C_Y} + \dot{\gamma} \left(R - R_H - \left(R + \frac{R_H \kappa}{1 + \kappa} \tan^2 \delta \right) \frac{\tilde{\varphi}^2}{2} \right) \cos \delta. \end{aligned} \quad (3.10)$$

For small $\kappa = \frac{R_H}{R_T} \ll 1$, the last formulas can also be expanded in κ , keeping terms up to κ^2 , $\kappa\tilde{\varphi}$, $\tilde{\varphi}^2$ inclusive. Having performed the expansion and the necessary transformations, we obtain

$$V_{P_X} = \frac{R_S}{R_T} V_{C_X} - \Omega_Z R_H \tilde{\varphi} \left(1 + \frac{R - R_H}{R_T} \right) \tan \delta - R\dot{\varphi} - \dot{\gamma} \left(R - R_H - \frac{R_H \tilde{\varphi}^2}{2} \right) \sin \delta, \quad (3.11)$$

$$V_{P_\tau} = \frac{R_S}{R_T} V_{C_Y} + \dot{\gamma} \left(R - R_H - \frac{R_H \tilde{\varphi}^2}{2} \right) \cos \delta. \quad (3.12)$$

In the limit, when transitioning to motion on a plane ($R_T \rightarrow \infty$, $\lambda = \frac{R_S}{R_T} \rightarrow 1$, $\kappa \rightarrow 0$), formulas (3.9), (3.10) and (3.11), (3.12) for V_{P_X} , V_{P_τ} change to (2.10), (2.11).

In accordance with (3.9) and (3.10), in the absence of slipping $V_{P_X} = V_{P_\tau} = 0$ the rotational speeds of the wheel and the roller are

$$\dot{\gamma} = \frac{\lambda V_{C_Y}}{(R_H + R_T c\tilde{\varphi}) c\alpha c\delta + R_H s\alpha s\delta s\tilde{\varphi} - R_S c\delta c\tilde{\varphi}}, \quad (3.13)$$

$$\dot{\varphi} = \frac{\lambda}{R_S c\alpha - R_T} \left[V_{C_X} c\alpha + V_{C_Y} \frac{(1 + \kappa c\tilde{\varphi} - \lambda c\alpha) s\delta + \lambda s\alpha c\delta s\tilde{\varphi}}{(\kappa + c\tilde{\varphi}) c\alpha c\delta + \kappa s\alpha s\delta s\tilde{\varphi} - \lambda c\delta c\tilde{\varphi}} - R_T \Omega_Z s\alpha \right]. \quad (3.14)$$

Substituting Eqs. (3.7) and (3.8) for α into the last expressions gives (with an accuracy of $\tilde{\varphi}^3$):

$$\dot{\varphi} = \frac{\lambda}{R} \left[\left(1 + \frac{R_H}{R} \frac{\tilde{\varphi}^2}{2} \frac{\kappa \tan^2 \delta}{(1 + \kappa)^2} \right) V_{C_X} + \left(1 + \frac{\tilde{\varphi}^2}{2} \frac{1 + \frac{R_H}{R} \kappa \tan^2 \delta - \kappa^2}{(1 + \kappa)^2} \right) V_{C_Y} \tan \delta - \frac{R_H}{1 + \kappa} \Omega_Z \tilde{\varphi} \tan \delta \right], \quad (3.15)$$

$$\dot{\gamma} = -\frac{\lambda V_{C_Y}}{(R - R_H) \cos \delta} \left[1 + \frac{\tilde{\varphi}^2}{2} \frac{R(1 + \kappa) + R_H \kappa \tan^2 \delta}{(R - R_H)(1 + \kappa)} \right]. \quad (3.16)$$

For $\kappa = \frac{R_H}{R_T} \ll 1$, the formulas for $\dot{\varphi}$ and $\dot{\gamma}$ can also be expanded in κ , keeping the terms up to κ^2 , $\kappa\tilde{\varphi}$, $\tilde{\varphi}^2$ inclusive:

$$\dot{\varphi} = \frac{1}{R} \left[\frac{R_S}{R_T} \left(V_{C_X} + V_{C_Y} \left(1 + \frac{R_T}{R_S} \frac{\tilde{\varphi}^2}{2} \right) \tan \delta \right) - \Omega_Z R_H \tilde{\varphi} \left(1 + \frac{R - R_H}{R_T} \right) \tan \delta \right], \quad (3.17)$$

$$\dot{\gamma} = -\frac{V_{C_Y}}{(R - R_H) \cos \delta} \left(\frac{R_S}{R_T} + \frac{R}{R - R_H} \frac{\tilde{\varphi}^2}{2} \right). \quad (3.18)$$

REMARK 4. In the absence of a displacement of the roller contact point at $\tilde{\varphi} = 0$, the formula for $\dot{\varphi}$ takes the form $R\dot{\varphi} = \frac{R_S}{R_T} (V_{C_X} + V_{C_Y} \tan \delta)$, which is obtained for the disk sliding along the sphere in the direction perpendicular to the roller axis [6].

Let us express the angular velocity of the contacting roller as the following sum:

$$\boldsymbol{\Omega}_r = \boldsymbol{\Omega}_{cf} + \dot{\varphi} \mathbf{e}_Y + \dot{\gamma} \mathbf{e}_r = \begin{pmatrix} \frac{V_{C_Y}}{R_T} + \dot{\gamma} \cos \delta \cos \tilde{\varphi} \\ -\frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta \\ \Omega_Z - \dot{\gamma} \cos \delta \sin \tilde{\varphi} \end{pmatrix}.$$

The rolling angular velocities of the roller are

$$\Omega_{r_X} = \frac{V_{C_Y}}{R_T} + \dot{\gamma} \cos \delta \cos \tilde{\varphi},$$

$$\Omega_{r\tau} = \mathbf{\Omega}_r \cdot \boldsymbol{\tau} = \left(-\frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta \right) \cos \alpha + (\Omega_Z - \dot{\gamma} \cos \delta \sin \tilde{\varphi}) \sin \alpha.$$

The spinning angular velocity of the roller is

$$\Omega_{rn} = \mathbf{\Omega}_r \cdot \mathbf{n} = - \left(-\frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta \right) \sin \alpha + (\Omega_Z - \dot{\gamma} \cos \delta \sin \tilde{\varphi}) \cos \alpha.$$

Let us present the expansion of the components of the angular velocity of the roller with respect to the angle $\tilde{\varphi}$:

$$\Omega_{r_X} = \frac{V_{C_Y}}{R_T} + \dot{\gamma} \left(1 - \frac{\tilde{\varphi}^2}{2} \right) \cos \delta, \quad (3.19)$$

$$\Omega_{r\tau} = \left(1 - \frac{\kappa^2}{(1+\kappa)^2} \frac{\tilde{\varphi}^2}{2} \tan^2 \delta \right) \left(\dot{\varphi} - \frac{V_{C_X}}{R_T} \right) + \frac{\kappa}{1+\kappa} \Omega_Z \tilde{\varphi} \tan \delta +$$

$$+ \dot{\gamma} \left(1 - \frac{\tilde{\varphi}^2}{2} \frac{(2 + \tan^2 \delta) \kappa^2 + 2\kappa}{(1+\kappa)^2} \right) \sin \delta, \quad (3.20)$$

$$\Omega_{rn} = \frac{\kappa \tilde{\varphi} \tan \delta}{1+\kappa} \left(\frac{V_{C_X}}{R_T} - \dot{\varphi} \right) + \Omega_Z \left(1 - \frac{\tilde{\varphi}^2}{2} \frac{\kappa^2 \tan^2 \delta}{(1+\kappa)^2} \right) - \dot{\gamma} \left(1 + \frac{\kappa \tan^2 \delta}{1+\kappa} \right) \tilde{\varphi} \cos \delta. \quad (3.21)$$

For $\kappa = \frac{R_H}{R_T} \ll 1$, the last two relations can also be expanded in κ , keeping terms up to κ^2 , $\kappa \tilde{\varphi}$, $\tilde{\varphi}^2$ inclusive:

$$\Omega_{r\tau} = -\frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta + \frac{R_H}{R_T} \Omega_Z \tilde{\varphi} \tan \delta, \quad (3.22)$$

$$\Omega_{rn} = \Omega_Z - \tilde{\varphi} \left(\dot{\gamma} \cos \delta + \frac{R_H}{R_T} \left(\dot{\varphi} - \frac{V_{C_X}}{R_T} + \dot{\gamma} \sin \delta \right) \tan \delta \right). \quad (3.23)$$

4. Kinematics of the Mecanum wheel on the external surface of a sphere

Let us turn our attention to the motion of the Mecanum wheel on the external surface of a spherical shell. In this case, the geometry of the rollers differs from those discussed earlier.

The set of rollers of an ideal Mecanum wheel for moving on the external surface of the sphere is enveloped by the surface of a ring-shaped torus formed by rotation of an arc of the sphere about the CY -axis of the wheel (Fig. 6). This enables the wheel to roll and spin, maintaining a constant distance from the center C to the supporting surface and a constant orientation of the CZ -axis, and during motion the contact point of the roller P is displaced relative to C only in the direction of the CY -axis. To describe the displacement, we introduce the angle $\alpha = \angle(\vec{OC}, \vec{OP})$ (see Fig. 7).

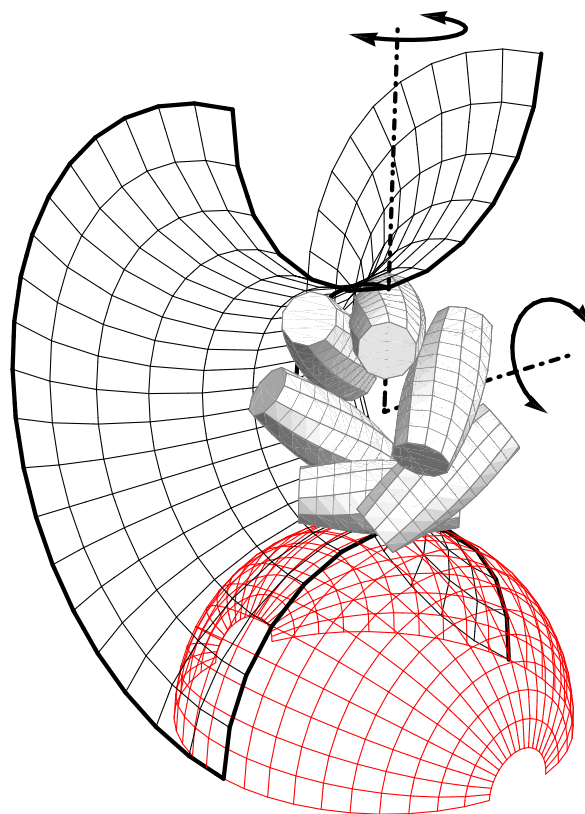


Fig. 6. Toric envelope of the Mecanum wheel rollers on the external surface of a sphere (the roller “meridians” are the lines of contact between the roller surface and the envelope surface; the possible directions and axes of the wheel rotation relative to the sphere are shown)

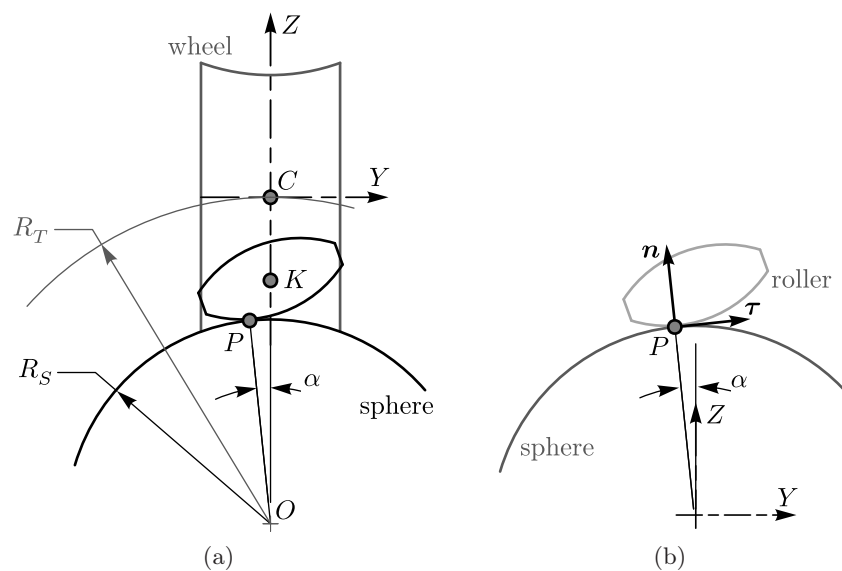


Fig. 7. Scheme of the external wheel on a sphere

Let R_S denote the radius of the spherical shell outside which the wheel moves (Fig. 7), and let $R_T = OC = \text{const}$ be the distance from the center of the wheel to the center of the sphere. Call their difference

$$R = R_T - R_S$$

the *radius of the wheel*.

Introduce the radius ratios

$$\lambda = \frac{R_S}{R_T}, \quad \kappa = \frac{R_H}{R_T},$$

where $R_H = CK$ is the wheel hub “radius”. Here $0 < \lambda < 1$, $\kappa > 0$.

Consider the kinematics of the wheel motion relative to the sphere.

The velocity at the contact point P relative to the spherical shell has the form

$$\mathbf{V}_P = \mathbf{V}_C + \boldsymbol{\Omega}_{\text{cf}} \times \overrightarrow{CP} + \dot{\varphi} \mathbf{e}_Y \times \overrightarrow{CP} + \dot{\gamma} \mathbf{e}_r \times \overrightarrow{KP}. \quad (4.1)$$

Here $\mathbf{V}_C = (V_{C_X} \ V_{C_Y} \ 0)^\top$ is the relative velocity of the wheel center C ; $\boldsymbol{\Omega}_{\text{cf}} = (-\frac{V_{C_Y}}{R_T} \ \frac{V_{C_X}}{R_T} \ \Omega_Z)^\top$ is the angular velocity of the XYZ -frame relative the sphere:

$$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \begin{pmatrix} 0 \\ -R_S \sin \alpha \\ R_S \cos \alpha \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ R_T \end{pmatrix} = \begin{pmatrix} 0 \\ -R_S \sin \alpha \\ R_S \cos \alpha - R_T \end{pmatrix}, \quad (4.2)$$

$$\overrightarrow{KP} = \overrightarrow{CP} - \overrightarrow{CK} = \begin{pmatrix} R_H \sin \tilde{\varphi} \\ -R_S \sin \alpha \\ R_H \cos \tilde{\varphi} + R_S \cos \alpha - R_T \end{pmatrix}, \quad (4.3)$$

the vectors $\overrightarrow{CK}$ and $\mathbf{e}_r$ are defined by the formulas (2.2) and (2.3).

Comparison of formulas (3.1)–(3.3) with similar ones (4.1)–(4.3) shows that all kinematic formulas obtained in Section 3 for the wheel inside the sphere can be extended to the case of the external wheel by a formal substitution

$$R_S \rightarrow -R_S, \quad R_T \rightarrow -R_T, \quad \kappa \rightarrow -\kappa.$$

Below we present only the main results on the kinematics of the Mecanum wheel on the sphere.

The velocity components of the contact point, P , have the form

$$\begin{aligned} V_{P_X} &= \frac{R_S}{R_T} V_{C_X} \text{c}\alpha + R_S \Omega_Z \text{s}\alpha - \\ &\quad - \dot{\varphi} (R_T - R_S \text{c}\alpha) - \dot{\gamma} ((R_T - R_H \text{c}\tilde{\varphi} - R_S \text{c}\alpha) \text{s}\delta + R_S \text{s}\alpha \text{c}\delta \text{s}\tilde{\varphi}), \\ V_{P_Y} &= \frac{R_S}{R_T} V_{C_Y} \text{c}\alpha + \dot{\gamma} ((R_T - R_S \text{c}\alpha) \text{c}\tilde{\varphi} - R_H) \text{c}\delta, \\ V_{P_Z} &= \frac{R_S}{R_T} V_{C_Y} \text{s}\alpha - \dot{\gamma} (R_H \text{s}\delta \text{s}\tilde{\varphi} + R_S \text{s}\alpha \text{c}\delta \text{c}\tilde{\varphi}). \end{aligned} \quad (4.4)$$

To analyze the conditions of contact between the rollers and the sphere, projections of relative velocities onto the normal $\mathbf{n} = \begin{pmatrix} 0 & -\sin \alpha & \cos \alpha \end{pmatrix}^\top$ and tangents X , $\boldsymbol{\tau} = \begin{pmatrix} 0 & \cos \alpha & \sin \alpha \end{pmatrix}^\top$ (Fig. 7) are used.

Project the velocity $\mathbf{V}_P$ onto the tangents and the normal to the sphere. The tangent component in the YZ -plane is

$$V_{P\tau} = \mathbf{V}_P \cdot \boldsymbol{\tau} = \frac{R_S}{R_T} V_{C_Y} - \dot{\gamma}((R_H - R_T c\tilde{\varphi}) c\alpha c\delta + R_H s\alpha s\delta s\tilde{\varphi} + R_S c\delta c\tilde{\varphi}). \quad (4.5)$$

The normal component of the contact point velocity is

$$V_{Pn} = \mathbf{V}_P \cdot \mathbf{n} = -\dot{\gamma}((R_T c\tilde{\varphi} - R_H) s\alpha c\delta + R_H s\tilde{\varphi} c\alpha s\delta).$$

From the condition of continuous contact between the wheel and the sphere at $V_{Pn} = 0$ we obtain

$$\begin{aligned} \tan \alpha &= -\frac{R_H \tan \delta \sin \tilde{\varphi}}{R_T \cos \tilde{\varphi} - R_H}, \\ R_T \tan \alpha &= -\frac{R_H \tan \delta \tan \tilde{\varphi}}{1 - \kappa \sec \tilde{\varphi}}. \end{aligned} \quad (4.6)$$

As $R_S \rightarrow \infty$ the sphere degenerates into a plane, $R_T \tan \alpha \rightarrow -\Delta_Y$ and $\kappa \rightarrow 0$. The shape of the wheel rollers in the limiting case corresponds to the results obtained above: for $R_S \rightarrow \infty$ the formula takes the form

$$\Delta_Y = -R_T \tan \alpha = R_H \tan \delta \tan \tilde{\varphi}.$$

Using (4.6), we find

$$\cos \alpha = \frac{1}{\sqrt{1 + \tan^2 \alpha}} = \frac{1 - \kappa \cos \tilde{\varphi}}{\sqrt{(1 - \kappa \cos \tilde{\varphi})^2 + (\kappa \tan \delta \sin \tilde{\varphi})^2}}, \quad (4.7)$$

$$\sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}} = \frac{-\kappa \tan \delta \sin \tilde{\varphi}}{\sqrt{(1 - \kappa \cos \tilde{\varphi})^2 + (\kappa \tan \delta \sin \tilde{\varphi})^2}} \quad (4.8)$$

for substitution into the subsequent kinematic relationships.

Substituting (4.7) and (4.8) into formulas (4.4) and (4.5) gives cumbersome and vast relations for V_{P_X} , $V_{P\tau}$. Let us expand them into a series in terms of the parameters κ and $\tilde{\varphi}$ up to cubic terms:

$$\begin{aligned} V_{P_X} &= \frac{R_S}{R_T} V_{C_X} - \Omega_Z R_H \tilde{\varphi} \left(1 - \frac{R - R_H}{R_T} \right) \tan \delta - R\dot{\varphi} - \dot{\gamma} \left(R - R_H - \frac{1}{2} R_H \tilde{\varphi}^2 \right) \sin \delta, \\ V_{P\tau} &= \frac{R_S}{R_T} V_{C_Y} + \dot{\gamma} \left(R - R_H - \frac{1}{2} R_H \tilde{\varphi}^2 \right) \cos \delta. \end{aligned}$$

In the absence of slipping $V_{P_X} = V_{P\tau} = 0$ from (4.4)–(4.8) we express the rotation velocities of the wheel and roller:

$$\dot{\varphi} = \frac{1}{R} \left[\frac{R_S}{R_T} \left(V_{C_X} + V_{C_Y} \left(1 + \frac{R_T}{R_S} \frac{\tilde{\varphi}^2}{2} \right) \tan \delta \right) - \Omega_Z R_H \tilde{\varphi} \left(1 - \frac{R - R_H}{R_T} \right) \tan \delta \right], \quad (4.9)$$

$$\dot{\gamma} = -\frac{V_{C_Y}}{(R - R_H) \cos \delta} \left(\frac{R_S}{R_T} + \frac{R}{R - R_H} \frac{\tilde{\varphi}^2}{2} \right). \quad (4.10)$$

These relations are expanded in terms of the parameters κ and $\tilde{\varphi}$ up to cubic terms.

Concluding the study of the kinematics of the Mecanum wheel on the sphere, we consider the components of the relative angular velocity of the roller $\mathbf{\Omega}_r = \mathbf{\Omega}_{cf} + \dot{\varphi}\mathbf{e}_Y + \dot{\gamma}\mathbf{e}_r$.

The rolling angular velocities of the roller are

$$\begin{aligned}\Omega_{r_X} &= \mathbf{\Omega}_r \cdot \mathbf{e}_X = -\frac{V_{C_Y}}{R_T} + \dot{\gamma} \cos \delta \cos \tilde{\varphi}, \\ \Omega_{r_\tau} &= \mathbf{\Omega}_r \cdot \boldsymbol{\tau} = \left(\frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta \right) \cos \alpha + (\Omega_Z - \dot{\gamma} \cos \delta \sin \tilde{\varphi}) \sin \alpha.\end{aligned}$$

The spinning angular velocity of the roller is

$$\Omega_{rn} = \mathbf{\Omega}_r \cdot \mathbf{n} = -\left(\frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta \right) \sin \alpha + (\Omega_Z - \dot{\gamma} \cos \delta \sin \tilde{\varphi}) \cos \alpha.$$

The expansions in terms of parameters κ and $\tilde{\varphi}$, up to cubic terms, have the form

$$\begin{aligned}\Omega_{r_X} &= -\frac{V_{C_Y}}{R_T} + \dot{\gamma} \left(1 - \frac{\tilde{\varphi}^2}{2} \right) \cos \delta, & \Omega_{r_\tau} &= \frac{V_{C_X}}{R_T} + \dot{\varphi} + \dot{\gamma} \sin \delta - \frac{R_H}{R_T} \Omega_Z \tilde{\varphi} \tan \delta, \\ \Omega_{rn} &= \Omega_Z - \tilde{\varphi} \left(\dot{\gamma} \cos \delta - \frac{R_H}{R_T} (\dot{\varphi} + \dot{\gamma} \sin \delta) \tan \delta \right).\end{aligned}$$

5. Geometry of the rollers of the Mecanum wheels

Using the geometric relations obtained above, we describe the shape of the roller of the ideal Mecanum wheel.

Let us obtain the equations of the roller surface. We assume that the generatrix considered lies in the $K\zeta\xi$ -plane, where the $K\zeta$ -axis is directed along the roller symmetry axis (with the unit vector $\mathbf{e}_r$), and $K\xi$ is perpendicular to it.

Find the coordinates of the contact point P in the axes of the roller. To do this, we decompose $\overrightarrow{KP}$ into components parallel and perpendicular to $\mathbf{e}_r$ and find their lengths ζ and ξ , respectively:

$$\zeta = \overrightarrow{KP} \cdot \mathbf{e}_r, \quad \xi = |\mathbf{e}_r \times [\overrightarrow{KP} \times \mathbf{e}_r]|.$$

The dependence $\xi(\zeta)$ expresses the change in the radius of the cross-section of the roller along the length of its axis.

Let us obtain the following equations of the generatrix in parametric form.

For the Mecanum wheel on the plane:

$$\zeta^{\text{pln}} = R c \delta s \tilde{\varphi} + \Delta_Y s \delta = R \cos \delta \sin \tilde{\varphi} + R_H \tan \delta \sin \delta \tan \tilde{\varphi}, \quad (5.1)$$

$$\xi^{\text{pln}} = \sqrt{(\Delta_Y c \delta - R s \delta s \tilde{\varphi})^2 + (R_H - R c \tilde{\varphi})^2} = (R \cos \tilde{\varphi} - R_H) \sqrt{\sin^2 \delta \tan^2 \tilde{\varphi} + 1}. \quad (5.2)$$

For the Mecanum wheel on the internal surface of the sphere:

$$\begin{aligned}\zeta^{\text{int}} &= (R_S c \alpha - R_T) c \delta s \tilde{\varphi} + R_S s \alpha s \delta, \\ \xi^{\text{int}} &= \sqrt{(R_S s \alpha c \delta - (R_S c \alpha - R_T) s \delta c \tilde{\varphi})^2 + (R_H - (R_S c \alpha - R_T) c \tilde{\varphi})^2},\end{aligned}$$

where $\cos \alpha$ and $\sin \alpha$ are defined by Eqs. (3.7) and (3.8).

For the Mecanum wheel on the external surface of the sphere:

$$\zeta^{\text{ext}} = (R_T - R_S \cos \alpha) c \delta s \tilde{\varphi} - R_S s \alpha s \delta, \quad (5.3)$$

$$\xi^{\text{ext}} = \sqrt{(R_S s \alpha c \delta + (R_T - R_S \cos \alpha) s \delta s \tilde{\varphi})^2 + (R_H - (R_T - R_S \cos \alpha) c \tilde{\varphi})^2}, \quad (5.4)$$

where $\cos \alpha$ and $\sin \alpha$ are defined by Eqs. (4.7) and (4.8).

Consider the asymptotic formula for the dependence of the radius of the cross section of the roller $\xi(\zeta)$, obtained by decomposition of degrees ζ :

$$\xi = R - R_H + \frac{D_2 \zeta^2}{2!} + \frac{D_4 \zeta^4}{4!}, \quad |\zeta| \leq \zeta_m,$$

where D_k are coefficients defined by the formula

$$D_k = \left(\frac{d^k \xi}{d\zeta^k} \right)_{\zeta=0} = \left(\frac{d^k \xi}{d\zeta^k} \right)_{\tilde{\varphi}=0}, \quad k = 2, 4,$$

using the rule of implicit differentiation

$$\frac{d}{d\zeta} = \frac{\frac{d}{d\tilde{\varphi}}}{\frac{d\zeta}{d\tilde{\varphi}}},$$

and $\zeta_m = \zeta|_{\tilde{\varphi}=\pi/N}$ is the length of the roller semiaxis.

Let us write out the values of D_k and ζ_m at $|\delta| = 45^\circ$:

- for the Mecanum wheel on the plane:

$$D_2^{\text{pln}} = -\frac{1}{R + R_H}, \quad D_4^{\text{pln}} = -3 \cdot \frac{R - R_H}{(R + R_H)^4},$$

$$\zeta_m^{\text{pln}} = \frac{R + R_H}{\sqrt{2}} \tilde{\varphi} + \frac{2R_H - R}{\sqrt{2}} \frac{\tilde{\varphi}^3}{6}, \quad \tilde{\varphi} = \frac{\pi}{N};$$

- for the Mecanum wheel on the internal surface of the sphere:

$$D_2^{\text{int}} = -\frac{1 + 2\kappa}{R + R_H + 2R\kappa},$$

$$D_4^{\text{int}} = -3 \cdot \frac{(1 + 2\kappa)^4 R + (12\kappa^3 + 16\kappa^2 + 4\kappa - 1) R_H}{(R + R_H + 2R\kappa)^4},$$

$$\zeta_m^{\text{int}} = \frac{R + R_H + 2R\kappa}{\sqrt{2}(1 + \kappa)} \tilde{\varphi} + \frac{(2 - 2\kappa - 7\kappa^2) R_H - (8\kappa^3 + 5\kappa^2 + \kappa + 1) R}{\sqrt{2}(1 + \kappa)^3} \frac{\tilde{\varphi}^3}{6}, \quad \tilde{\varphi} = \frac{\pi}{N};$$

- for the Mecanum wheel on the external surface of the sphere:

$$D_2^{\text{ext}} = -\frac{1 - 2\kappa}{R + R_H - 2R\kappa},$$

$$D_4^{\text{ext}} = -3 \cdot \frac{(1 - 2\kappa)^4 R - (12\kappa^3 - 16\kappa^2 + 4\kappa + 1) R_H}{(R + R_H - 2R\kappa)^4},$$

$$\zeta_m^{\text{ext}} = \frac{R + R_H - 2R\kappa}{\sqrt{2}(1 - \kappa)} \tilde{\varphi} + \frac{(2 + 2\kappa - 7\kappa^2) R_H + (8\kappa^3 - 5\kappa^2 + \kappa - 1) R}{\sqrt{2}(1 - \kappa)^3} \frac{\tilde{\varphi}^3}{6}, \quad \tilde{\varphi} = \frac{\pi}{N}.$$

With an arbitrary δ , the expressions for D_k and ζ_m are more cumbersome. For example, for the wheel inside the sphere we get

$$\begin{aligned}
 D_2^{\text{int}} &= -\frac{1 + 2\kappa + \cos 2\delta}{R + R_H + 2R\kappa + (R - R_H) \cos 2\delta}, \\
 D_4^{\text{int}} &= -\frac{3((128\kappa^4 + 256\kappa^3 + 288\kappa^2 + 160\kappa + 35)R + (80\kappa^3 + 64\kappa^2 + 16\kappa - 3)R_H)}{8(R + R_H + 2R\kappa + (R - R_H) \cos 2\delta)^4} - \\
 &\quad - \frac{24((32\kappa^3 + 48\kappa^2 + 30\kappa + 7)R + (\kappa - 8\kappa^3)R_H)}{8(R + R_H + 2R\kappa + (R - R_H) \cos 2\delta)^4} \cos 2\delta - \\
 &\quad - \frac{12((24\kappa^2 + 24\kappa + 7)R - (4\kappa^3 + 16\kappa^2 + 4\kappa - 1)R_H)}{8(R + R_H + 2R\kappa + (R - R_H) \cos 2\delta)^4} \cos 4\delta - \\
 &\quad - \frac{24(2R\kappa + R - R_H\kappa) \cos 6\delta + 3(R - R_H) \cos 8\delta}{8(R + R_H + 2R\kappa + (R - R_H) \cos 2\delta)^4}, \\
 \zeta_m^{\text{int}} &= \frac{R + R_H + 2R\kappa + (R - R_H) \cos 2\delta}{2(\kappa + 1) \cos \delta} \tilde{\varphi} - \\
 &\quad - \frac{(13\kappa^2 + 2\kappa - 2)R_H + (16\kappa^3 + 11\kappa^2 + 7\kappa + 3)R}{8(1 + \kappa)^3 \cos^3 \delta} \frac{\tilde{\varphi}^3}{6} + \\
 &\quad + \frac{3\kappa^2 R_H + (2\kappa^3 - 3\kappa^2 - 3\kappa - 1)R}{2(1 + \kappa)^3 \cos^3 \delta} \frac{\tilde{\varphi}^3}{6} \cos 2\delta + \\
 &\quad + \frac{(\kappa^2 + 2\kappa - 2)R_H - (\kappa^2 + 5\kappa + 1)R}{8(1 + \kappa)^3 \cos^3 \delta} \frac{\tilde{\varphi}^3}{6} \cos 4\delta.
 \end{aligned}$$

Similar expressions for the Mecanum wheels on the plane and outside the sphere can be obtained by the rule

$$\begin{aligned}
 D_k^{\text{ext}}(\kappa) &= D_k^{\text{int}}(-\kappa), \quad D_k^{\text{pln}} = D_k^{\text{int}}(\kappa = 0); \\
 \zeta_m^{\text{ext}}(\kappa) &= \zeta_m^{\text{int}}(-\kappa), \quad \zeta_m^{\text{pln}} = \zeta_m^{\text{int}}(\kappa = 0).
 \end{aligned}$$

Analysis of the relationship shows that for small values of $\kappa = \frac{R_H}{R_T} \ll 1$

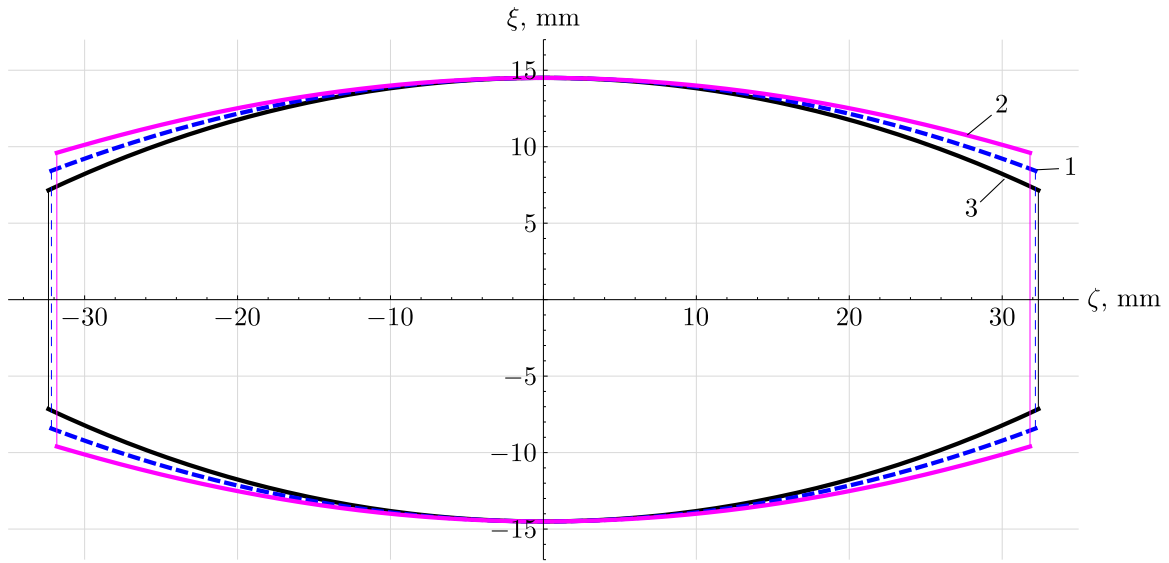
$$\begin{aligned}
 |D_k^{\text{ext}}| &< |D_k^{\text{pln}}| < |D_k^{\text{int}}|, \\
 \zeta_m^{\text{ext}} &< \zeta_m^{\text{pln}} < \zeta_m^{\text{int}}.
 \end{aligned}$$

Thus, the roller of the Mecanum wheel for the internal surface of the sphere is slightly longer than the others and has more curvature of the generatrix. The roller for the outer surface of the sphere is slightly shorter than the others and has less curvature of the generatrix.

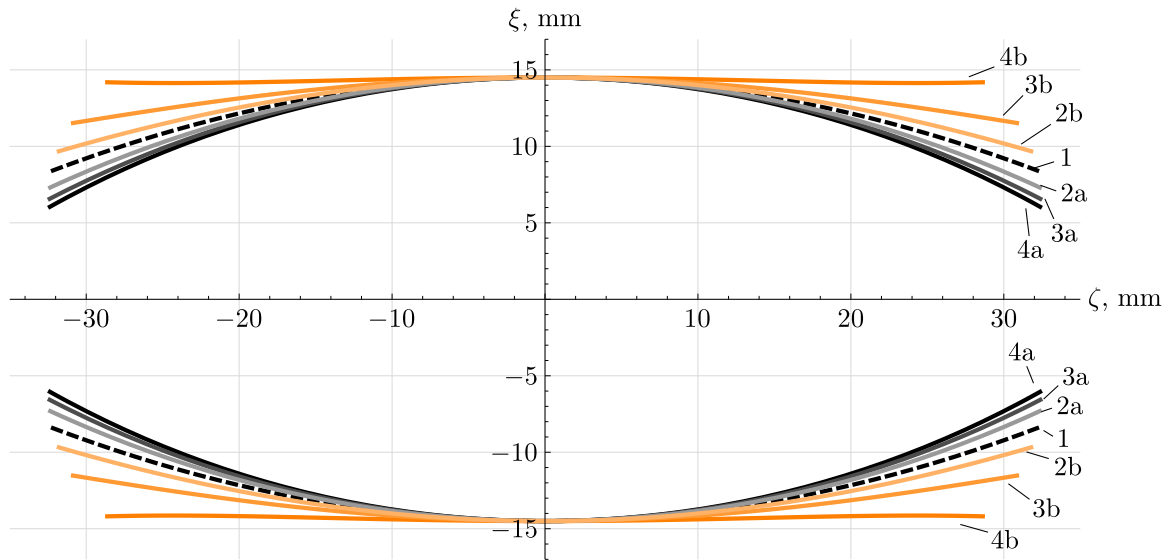
Figure 8 shows the shapes of the rollers and graphs of their generatrices for a Mecanum wheel with $N = 6$ rollers, radius $R = 50$ mm and hub radius $R_H = 35.5$ mm. The specified parameters are given for the wheels of the KUKA youBot [32] robot.

Note that, for $\kappa \approx 0.5$, the wheel roller generatrix for the external surface of the sphere differs slightly from the straight line (for $\kappa = 0.5$, the coefficient $D_2^{\text{ext}} = 0$, but at the same time $D_4^{\text{ext}} \neq 0$). For the rollers in Fig. 8b, this property is achieved at $\kappa = 0.45$ (curves 4b).

Figure 9 shows the positions of the Mecanum wheel rollers when rolling on a plane or a sphere. The shape of the rollers is in agreement with the results obtained above. Note that, while maintaining a constant distance to the support surface, the rollers do not go beyond it.



(a) Shapes of the rollers at $R_S = 200$ mm, $\kappa = 0.237$. 1 — wheel on a plane; 2 — wheel outside a sphere; 3 — wheel inside a sphere



(b) Shapes of the rollers at different radii of the sphere. 1 — wheel on a plane; wheel inside a sphere: 2a — $R_S = 228$ mm, $\kappa = 0.2$; 3a — $R_S = 139$ mm, $\kappa = 0.4$; 4a — $R_S = 109$ mm, $\kappa = 0.6$; wheel outside a sphere: 2b — $R_S = 187$ mm, $\kappa = 0.15$; 3b — $R_S = 68$ mm, $\kappa = 0.3$; 4b — $R_S = 29$ mm, $\kappa = 0.45$

Fig. 8. Shapes of the Mecanum wheel rollers ($R = 50$ mm, $R_H = 35.5$ mm, $N = 6$)

At the end of this section, we note one property of the track of the Mecanum wheels on the xy plane (Fig. 10).

Consider the motion of the center of the wheel with velocities $\dot{x}_C = V_{C_x} = \text{const}$, $\dot{y}_C = V_{C_y} = 0$, $\Omega_Z = 0$. In the absence of slippage, the contacting rollers do not rotate, $\dot{\gamma} = 0$, and $R\dot{\varphi} = V_{C_x}$. The coordinates of the point of contact of the roller with the supporting plane are as follows:

$$x_P = R\varphi, \quad y_P = \Delta_Y = R_H \tan \delta \tan \tilde{\varphi}.$$

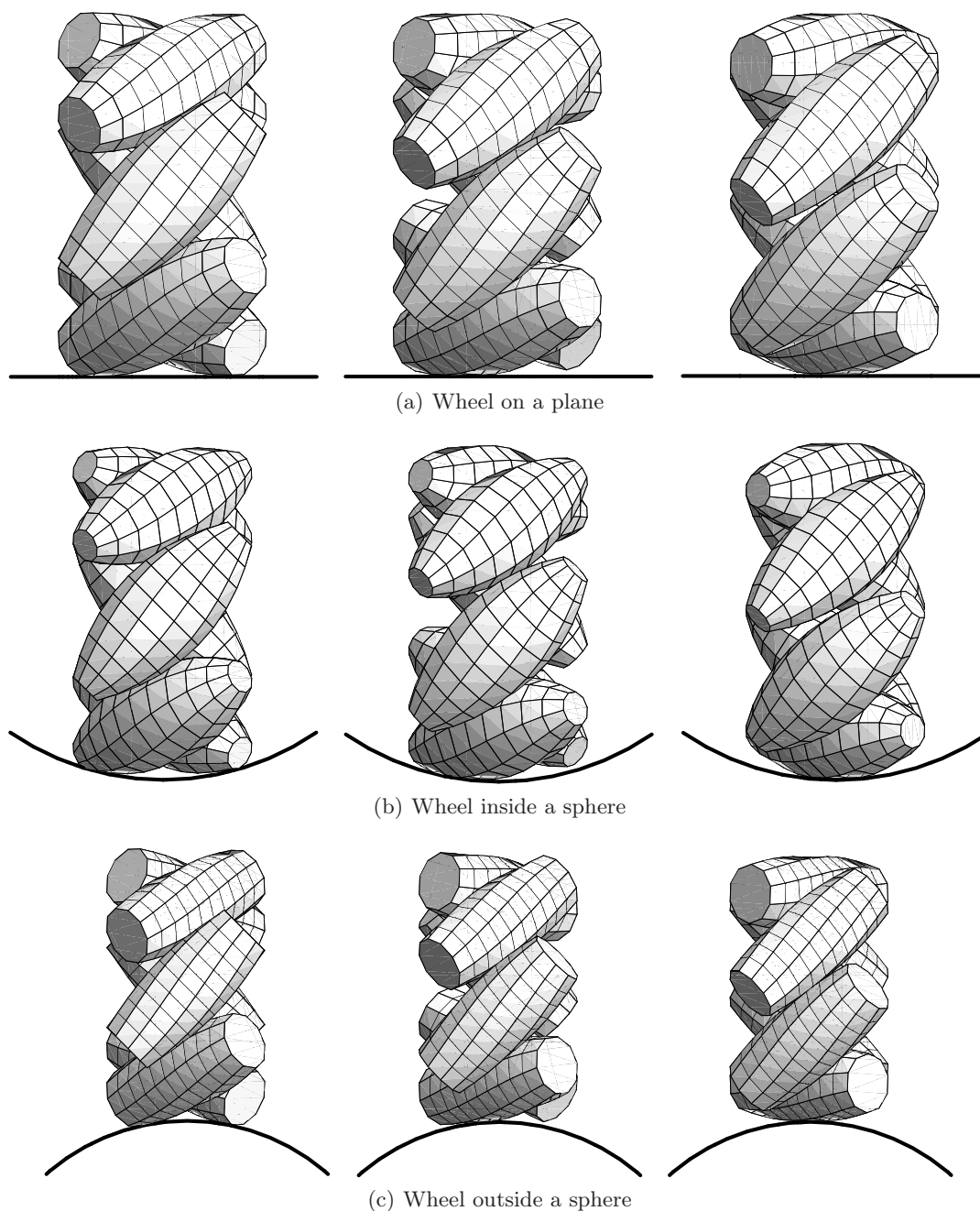


Fig. 9. Position of the rollers during the rolling of the Mecanum wheel on a plane or a sphere

The corresponding graph is shown in Fig. 10b for the values $N = 6$, $R = 50$ mm, $R_H = 35.5$ mm [32].

In Fig. 10a, the “meridians” of the roller are the geometric places of the contact points P at $V_{C_Y} = 0$, $\Omega_Z = 0$ at various initial angles γ (i. e., the tracks of the plane on the roller). In all other figures, the surfaces of the rollers are generated in a similar way.

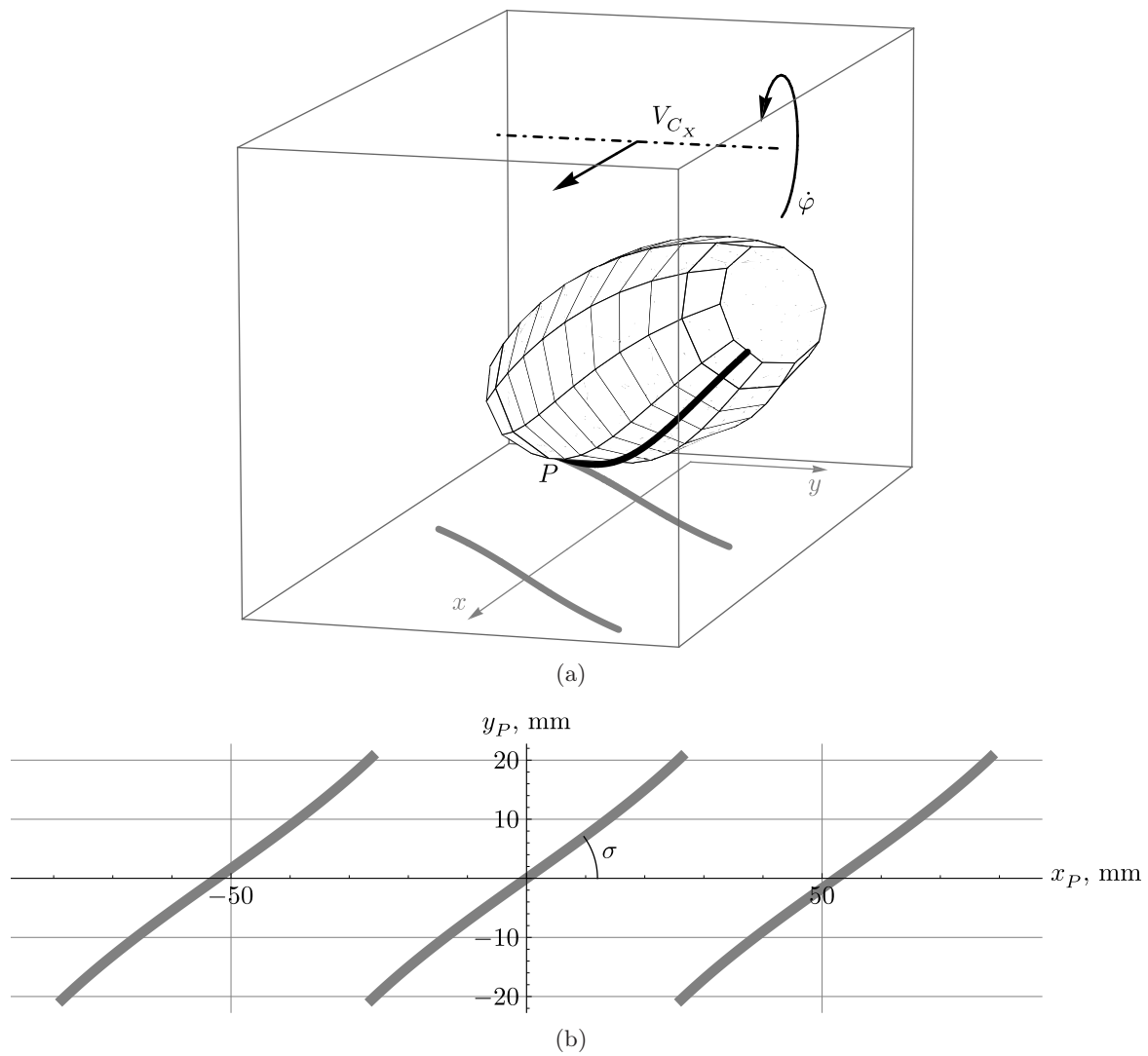


Fig. 10. Track of the Mecanum wheel when rolling on a plane

The average slope of the track to the x -axis (see Fig. 10) is

$$\tan \sigma = \left. \frac{y_P}{x_P} \right|_{\tilde{\varphi}=\pi/N} = \left. \frac{R_H \tan \delta \tan \tilde{\varphi}}{R \tilde{\varphi}} \right|_{\tilde{\varphi}=\pi/N} \approx \left(1 + \frac{\pi^2}{3N^2} \right) \frac{R_H}{R} \tan \delta.$$

Using this formula, the value of the angle δ or the hub radius R_H can be experimentally determined from the wheel track.

6. Example: a study of the kinematics of a spherical robot with an internal mecanum platform

6.1. A scheme and basic kinematic relationships

As an example, consider the motion of a spherical robot with an internal three-wheeled Mecanum platform.

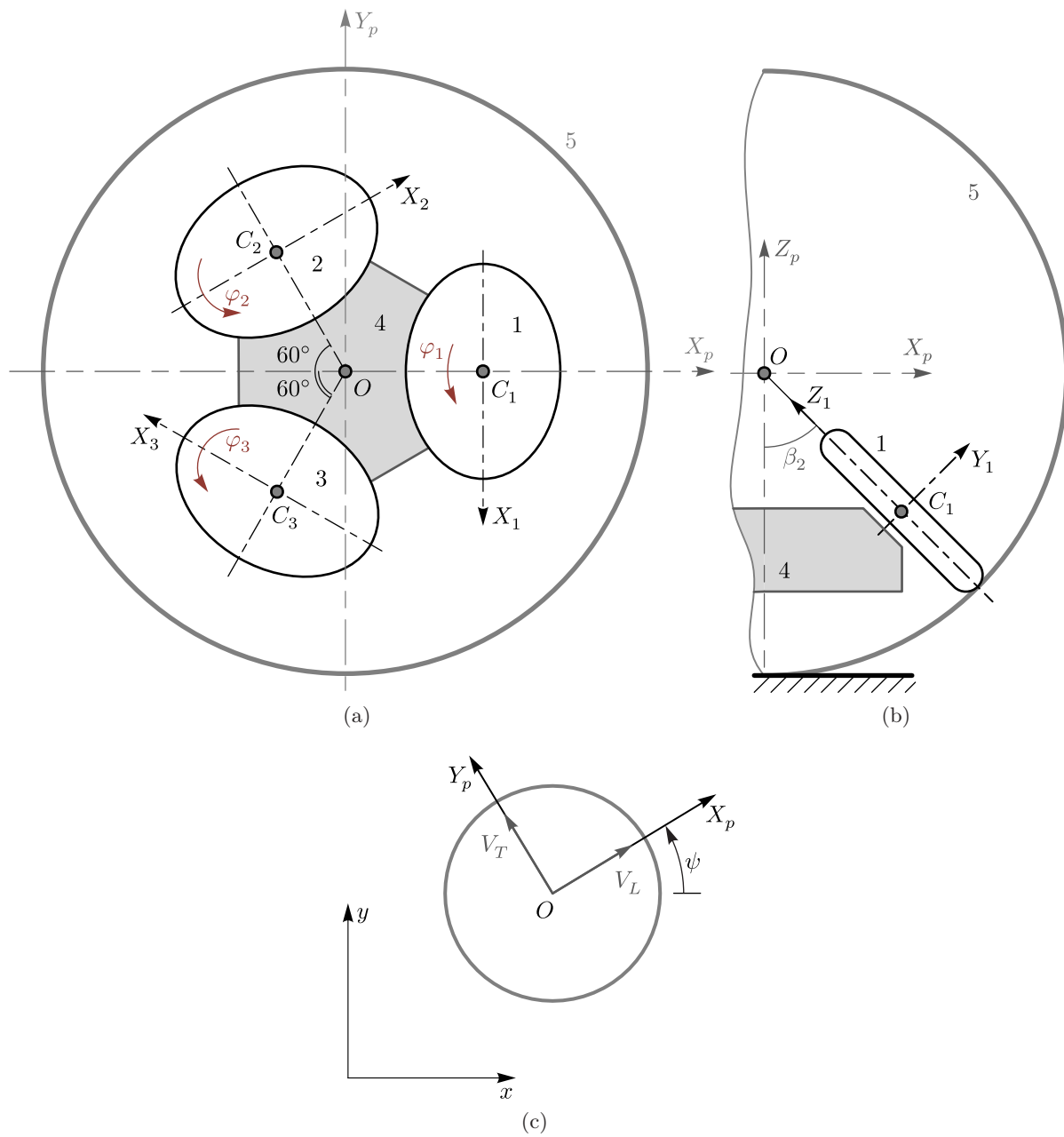


Fig. 11. Scheme of the spherical robot (1–3 — Mecanum wheels, 4 — platform, 5 — spherical shell)

A robot scheme is shown in Fig. 11. The internal platform 4 is driven by symmetrically located Mecanum wheels 1–3, oriented normal to the spherical shell 5. The spherobot moves along the horizontal plane xy .

The angles of deflection of the roller axes for all wheels are the same:

$$\delta_1 = \delta_2 = \delta_3 = \delta = 45^\circ.$$

Let $OX_pY_pZ_p$ be a coordinate frame of the platform 4 with origin at the center of the sphere O . Coordinate frames $C_iX_iY_iZ_i$, $i = 1, 2, 3$, with origins at the centers of the wheels C_i

are introduced similarly to the coordinate systems $CXYZ$ discussed above (the wheels rotate about the $C_i Y_i$ -axes). These coordinate frames are fixed relative to $O X_p Y_p Z_p$, their positions are described by the angles

$$\beta_{1i} = \angle(Z_p, Z_i), \quad \beta_{2i} = \beta_2 = \angle(X_p, X_i), \quad i = 1, 2, 3.$$

Here

$$\beta_{11} = -90^\circ, \quad \beta_{12} = 30^\circ, \quad \beta_{13} = 150^\circ, \quad \beta_2 = 45^\circ.$$

The transformation of vectors during the transition from $X_i Y_i Z_i$ to $X_p Y_p Z_p$ is carried out by a matrix of direction cosines

$$\Gamma_i = \begin{bmatrix} c\beta_{1i} & -s\beta_{1i} & 0 \\ s\beta_{1i} & c\beta_{1i} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\beta_2 & -s\beta_2 \\ 0 & s\beta_2 & c\beta_2 \end{bmatrix} = \begin{bmatrix} c\beta_{1i} & -s\beta_{1i} c\beta_2 & s\beta_{1i} s\beta_2 \\ s\beta_{1i} & c\beta_{1i} c\beta_2 & -c\beta_{1i} s\beta_2 \\ 0 & s\beta_2 & c\beta_2 \end{bmatrix}, \quad (6.1)$$

which is the product of the rotation matrices by the angles β_{1i} and β_2 .

The robot's motion is considered only at the kinematic level. The following assumptions are made:

- the spherical shell does not slip on the plane, there is no spinning;
- the wheel rollers do not slip on the sphere;
- the platform moves translationally, the plane $X_p Y_p$ is parallel to xy , the angle of rotation of the platform $\psi = \angle(x, X_p)$, see Fig. 11;
- the wheel drives are ideal: they precisely track the given desired wheel speeds $\dot{\varphi}_i^d$, $i = 1, 2, 3$.

Denote by V_L and V_T projections of the velocity of the sphere center O on the axes X_p and Y_p (Fig. 11):

$$V_L = \dot{x} \cos \psi + \dot{y} \sin \psi, \quad V_T = -\dot{x} \sin \psi + \dot{y} \cos \psi,$$

where x and y are the coordinates of the center O of the sphere.

The absolute angular velocity of the sphere and the platform in the $X_p Y_p Z_p$ -frame is

$$\omega_s^{(p)} = \left(-\frac{V_T}{R_S} \quad \frac{V_L}{R_S} \quad 0 \right)^\top, \quad \omega_p^{(p)} = \left(0 \quad 0 \quad \dot{\psi} \right)^\top,$$

where the superscript (p) indicates the platform in the $X_p Y_p Z_p$ -frame.

The angular velocity of the platform relative to the sphere in the $X_i Y_i Z_i$ -frame is

$$\Omega = \Gamma_i^\top \left(\omega_p^{(p)} - \omega_s^{(p)} \right) = \frac{1}{R_S} \begin{bmatrix} c\beta_{1i} & s\beta_{1i} & 0 \\ -s\beta_{1i} c\beta_2 & c\beta_{1i} c\beta_2 & s\beta_2 \\ s\beta_{1i} s\beta_2 & -c\beta_{1i} s\beta_2 & c\beta_2 \end{bmatrix} \begin{pmatrix} V_T \\ -V_L \\ \dot{\psi} R_S \end{pmatrix}.$$

We get

$$\begin{aligned} \Omega_{Xi} &= \frac{V_T}{R_S} c\beta_{1i} - \frac{V_L}{R_S} s\beta_{1i}, \\ \Omega_{Yi} &= -\frac{V_T}{R_S} s\beta_{1i} c\beta_2 - \frac{V_L}{R_S} c\beta_{1i} c\beta_2 + \dot{\psi} s\beta_2, \\ \Omega_{Zi} &= \frac{V_T}{R_S} s\beta_{1i} s\beta_2 + \frac{V_L}{R_S} c\beta_{1i} s\beta_2 + \dot{\psi} c\beta_2. \end{aligned}$$

The velocity of the wheel center C_i relative to the sphere is

$$\mathbf{V}_{C_i} = \boldsymbol{\Omega} \times \overrightarrow{OC_i},$$

where $\overrightarrow{OC_i} = \begin{pmatrix} 0 & 0 & -R_T \end{pmatrix}^\top$ in the $X_i Y_i Z_i$ -frame.

The components of the relative velocity of C_i are

$$\begin{aligned} V_{C_i X_i} &= -\Omega_{Y_i} R_T = \frac{R_T}{R_S} (V_T s\beta_{1i} c\beta_2 + V_L c\beta_{1i} c\beta_2) - R_T \dot{\psi} s\beta_2, \\ V_{C_i Y_i} &= \Omega_{X_i} R_T = \frac{R_T}{R_S} (V_T c\beta_{1i} - V_L s\beta_{1i}), \\ V_{C_i Z_i} &= 0. \end{aligned}$$

Substituting expressions for $V_{C_i X_i}$, $V_{C_i Y_i}$, Ω_{Z_i} instead of V_{C_X} , V_{C_Y} , Ω_Z into (3.14) gives “full-size” equations for determining the wheel speeds

$$\begin{aligned} \dot{\varphi}_i &= \omega_{Fi}(V_L, V_T, \dot{\psi}), \quad i = 1, 2, 3; \\ \dot{\varphi}_i &= \frac{\lambda}{R_S c\alpha_i - R_T} \left[V_{C_i X_i} c\alpha_i + V_{C_i Y_i} \frac{(1 + \kappa c\tilde{\varphi}_i - \lambda c\alpha_i) s\delta_i + \lambda s\alpha_i c\delta_i s\tilde{\varphi}_i}{(\kappa + c\tilde{\varphi}_i) c\alpha_i c\delta_i + \kappa s\alpha_i s\delta_i s\tilde{\varphi}_i - \lambda c\delta_i c\tilde{\varphi}_i} - R_T \Omega_{Z_i} s\alpha_i \right], \\ c\alpha_i &= \frac{1 + \kappa c\tilde{\varphi}_i}{\sqrt{(1 + \kappa c\tilde{\varphi}_i)^2 + (\kappa \tan \delta_i s\tilde{\varphi}_i)^2}}, \quad s\alpha_i = \frac{\kappa \tan \delta_i s\tilde{\varphi}_i}{\sqrt{(1 + \kappa c\tilde{\varphi}_i)^2 + (\kappa \tan \delta_i s\tilde{\varphi}_i)^2}}. \end{aligned} \quad (6.2)$$

A similar substitution into (3.17) gives “reduced” equations for determining the wheel speeds

$$\begin{aligned} \dot{\varphi}_i &= \omega_{Ri}(V_L, V_T, \dot{\psi}), \quad i = 1, 2, 3; \\ \dot{\varphi}_i &= \frac{1}{R} \left[\frac{R_S}{R_T} \left(V_{C_i X_i} + V_{C_i Y_i} \left(1 + \frac{R_T}{R_S} \frac{\tilde{\varphi}_i^2}{2} \right) \tan \delta_i \right) - \Omega_Z R_H \tilde{\varphi}_i \left(1 + \frac{R - R_H}{R_T} \right) \tan \delta_i \right]. \end{aligned} \quad (6.3)$$

6.2. Simulation methodology and initial data

An analysis of the accuracy of the spherical robot motion and the influence of the geometry of the Mecanum wheels on it is carried out as follows:

1. The desired motion is set — the required position of the robot center and the angle of rotation of the platform are specified:

$$\begin{aligned} x &= x^d(t), \quad y = y^d(t), \quad \psi = \psi^d(t), \\ V_L^d &= \dot{x}^d \cos \psi^d + \dot{y}^d \sin \psi^d, \quad V_T^d = -\dot{x}^d \sin \psi^d + \dot{y}^d \cos \psi^d. \end{aligned}$$

2. To determine the wheels' rotation velocities for the desired motion, kinematic equations derived without taking into account the displacements of the rollers' contact point and the sphere ($\alpha = 0$, $\tilde{\varphi} = 0$) are used:

$$\begin{aligned} \dot{\varphi}_i^d &= \omega_{Fi} \left(V_L^d, V_T^d, \dot{\psi}^d \right) \Big|_{\tilde{\varphi}=0}, \quad i = 1, 2, 3; \\ \dot{\varphi}_i^d &= \frac{1}{R} \left((c\beta_{1i} c\beta_2 - s\beta_{1i} \tan \delta_i) V_L^d + (s\beta_{1i} c\beta_2 + c\beta_{1i} \tan \delta_i) V_T^d - R_S \dot{\psi}^d s\beta_2 \right). \end{aligned}$$

3. The robot motion is determined by numerical solution of the kinematic equations

$$\begin{aligned} \dot{x} &= V_L \cos \psi - V_T \sin \psi, & \dot{y} &= V_L \sin \psi + V_T \cos \psi, \\ \omega_{Fi}(V_L, V_T, \dot{\psi}) &= \dot{\varphi}_i^d \quad \text{or} \quad \omega_{Ri}(V_L, V_T, \dot{\psi}) = \dot{\varphi}_i^d, & i &= 1, 2, 3. \end{aligned}$$

Consider several desired motions at $0 \leq t \leq t_f$ with trajectories representing a square with side 2ρ or a circle of radius ρ . Choose

$$\rho = 0.2 \text{ m}, \quad t_f = 4 \text{ s}.$$

The translational motion along the square is

$$\begin{cases} \dot{x}^d = 0, & \dot{y}^d = V^d, & \dot{\psi}^d = 0, & 0 \leq t \leq t_1, \\ \dot{x}^d = V^d, & \dot{y}^d = 0, & \dot{\psi}^d = 0, & t_1 < t \leq 2t_1, \\ \dot{x}^d = 0, & \dot{y}^d = -V^d, & \dot{\psi}^d = 0, & 2t_1 < t \leq 3t_1, \\ \dot{x}^d = -V^d, & \dot{y}^d = 0, & \dot{\psi}^d = 0, & 3t_1 < t \leq 4t_1 \end{cases} \quad (\text{A})$$

with $x^d(0) = y^d(0) = 0$, $V^d = \frac{2\rho}{t_1} = 0.4 \text{ m/s}$, $t_1 = 1 \text{ s}$.

The translational motion along the circle is

$$\dot{\psi}^d = 0, \quad V_L^d = \frac{2\pi\rho}{t_f} \sin \frac{2\pi t}{t_f}, \quad V_T^d = \frac{2\pi\rho}{t_f} \cos \frac{2\pi t}{t_f} \quad (\text{B})$$

with $x^d(0) = y^d(0) = 0$, $\psi^d(0) = 0$.

The longitudinal motion along the circle is

$$\dot{\psi}^d = -\frac{2\pi}{t_f}, \quad V_L^d = -\rho\dot{\psi}^d = \frac{2\pi\rho}{t_f}, \quad V_T^d = 0 \quad (\text{C})$$

with $x^d(0) = y^d(0) = 0$, $\psi^d(0) = \frac{\pi}{2}$.

The lateral motion along the circle is

$$\dot{\psi}^d = -\frac{2\pi}{t_f}, \quad V_L^d = 0, \quad V_T^d = -\rho\dot{\psi}^d = \frac{2\pi\rho}{t_f} \quad (\text{D})$$

with $x^d(0) = y^d(0) = 0$, $\psi^d(0) = 0$.

All of the specified trajectories are traversed clockwise, and their origin coincides with the origin of the coordinates xy .

Consider two sets of geometric parameters of the robot corresponding to different sizes of the wheels and the spherical shell:

$$R = 50 \text{ mm}, \quad R_H = 35.5 \text{ mm}, \quad R_S = 200 \text{ mm}, \quad \kappa = 0.237, \quad \lambda = 1.333; \quad (\text{I})$$

$$R = 70 \text{ mm}, \quad R_H = 55.5 \text{ mm}, \quad R_S = 170 \text{ mm}, \quad \kappa = 0.555, \quad \lambda = 1.7. \quad (\text{II})$$

The number of rollers on each wheel is $N = 6$ or $N = 8$.

When choosing parameters (I), the characteristics of the wheels [32] were used; the choice of parameters (II) and the laws of motions (A), (B) is based on [6].

6.3. Simulation results: the effect of the simplification of the kinematic equations

Using the example of motion (A), let us compare simulation results when using the “full-size” $\omega_{Fi}(V_L, V_T, \dot{\psi}) = \dot{\varphi}_i^d$ and the “reduced” $\omega_{Ri}(V_L, V_T, \dot{\psi}) = \dot{\varphi}_i^d$ kinematics equations for expressing the platform velocities $V_L, V_T, \dot{\psi}$.

Figure 12 shows graphs of the trajectories of the robot center and the angle of rotation of the platform with geometric parameters (I) and $N = 6$ rollers at $\varphi_1(0) = \varphi_2(0) = 0, \varphi_3(0) = 0.3$. The indicated graphs for the “full-size” and the “reduced” kinematics equations practically merge with each other. The velocity mismatch when using the specified kinematic relationships

$$\Delta_{R-F}(v_x) = \dot{x}|_{\omega_{Ri}=\dot{\varphi}_i^d} - \dot{x}|_{\omega_{Fi}=\dot{\varphi}_i^d}$$

is no more than $4 \cdot 10^{-4}$ m/s. On average $\Delta_{R-F}(v_x) \approx 10^{-4}$ m/s, which gives about $4 \cdot 10^{-4}$ m x -coordinate mismatch during the movement.

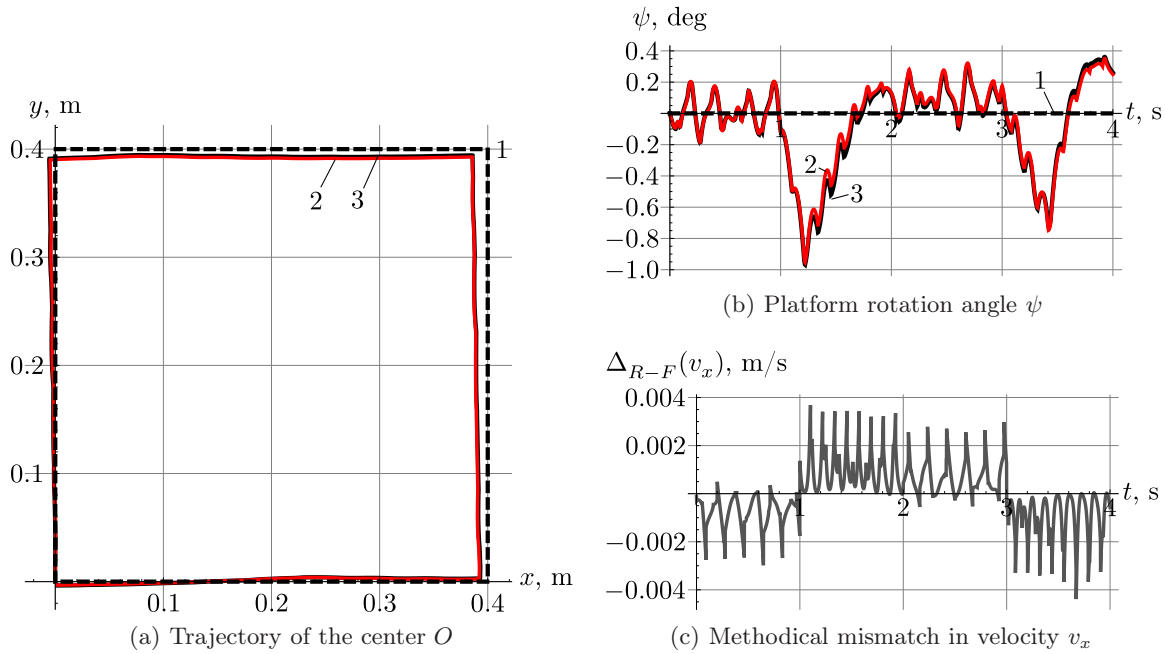


Fig. 12. Motion along the square: comparison of the kinematic equations (parameters (I), $N = 6$). 1 — desired motion (A); numerical simulation results using: 2 — the “full-size” Eqs. $\omega_{Fi} = \dot{\varphi}_i^d$; 3 — the “reduced” Eqs. $\omega_{Ri} = \dot{\varphi}_i^d$

At the same time, the kinematic errors of the motion tracking

$$\Delta v_x = \dot{x} - \dot{x}_d, \quad \Delta v_y = \dot{y} - \dot{y}_d, \quad \Delta x = x - x^d, \quad \Delta y = y - y^d$$

(Fig. 13) are greater than the specified methodological errors by two orders of magnitude:

$$\max |\Delta v_x| \approx \max |\Delta v_y| \approx 0.04 \text{ m/s}, \quad \max |\Delta x| = 14 \text{ mm}, \quad \max |\Delta y| = 9 \text{ mm}.$$

Thus, in the case under consideration, the simplification of the kinematics equations gives negligible methodological errors.

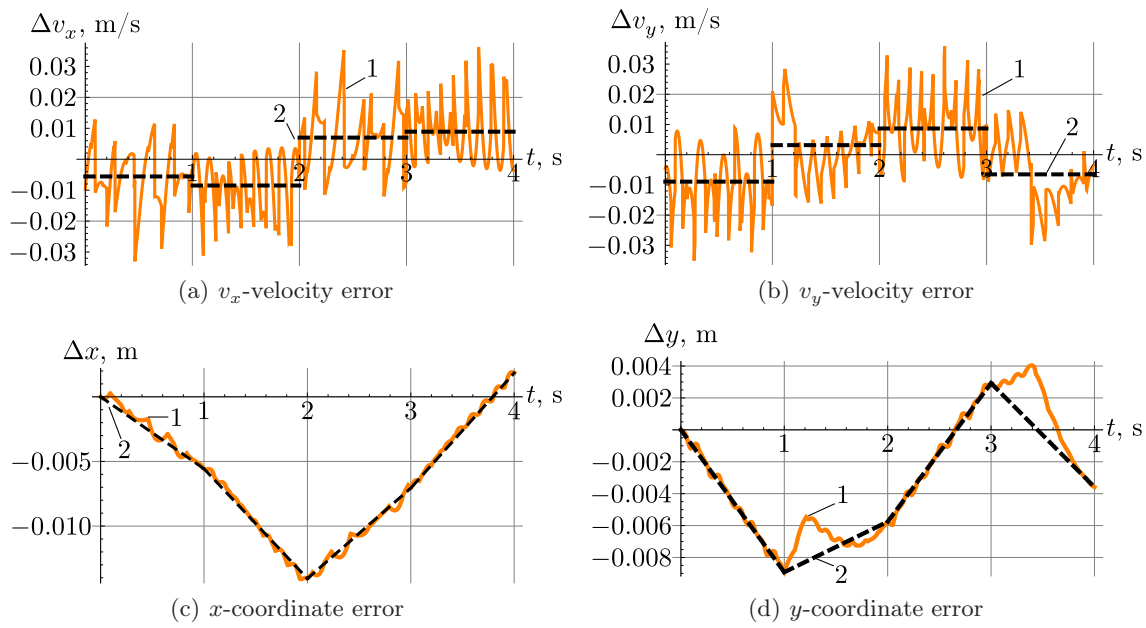


Fig. 13. Velocity and position errors when tracking motion (A) (parameters (I), $N = 6$). 1 — errors; 2 — average errors, trends

The change in errors Δv_x , Δv_y is of a high-frequency nature, caused by an abrupt change in $\tilde{\varphi}_i$ when changing the contacting rollers. Due to the nonzero average values Δv_x , Δv_y , caused primarily by the presence of $\tilde{\varphi}_i^2$ in the kinematic equations, the errors Δx , Δy have piecewise linear trends (Fig. 13), explaining the “drift” of the trajectory compared to the desired one.

Now let us consider numerical simulation results for the robot motion with geometric parameters (II), with $N = 6$ and $N = 8$ rollers. Graphs of trajectories of the center O , angles ψ , methodological velocity errors $\Delta_{R-F}(v_x)$ for the robot at $\varphi_1(0) = \varphi_2(0) = 0$, $\varphi_3(0) = 0.3$ are shown in Fig. 14.

For $N = 6$ rollers, the methodical mismatch in angle ψ reaches 0.5° , and the maximum mismatch in velocity is $\max |\Delta_{R-F}(v_x)| = 0.01$ m/s. At $N = 8$, the indicated errors decrease by a factor of several and correspond to the values obtained for the robot with parameters (I) (Fig. 12).

Further, the “reduced” equations are used in simulating the robot motion with parameters (I), since this does not lead to a loss of accuracy, but allows one to reduce computational costs; and for the case of parameters (II), the “full-size” equations are used.

6.4. Simulation results: analysis of the accuracy of motion

Let us consider the characteristics of the accuracy of tracking of motions (A)–(D).

The accuracy analysis is performed based on a set of simulation results obtained for an aggregate Φ_0 of 40 triplets of initial conditions $\{\varphi_1(0), \varphi_2(0), \varphi_3(0)\}$, where $\varphi_i(0) \in [-\frac{\pi}{6}, \frac{\pi}{6}]$ were generated as independent uniformly distributed random variables.

The following characteristics of the desired motion tracking accuracy are used below:

- a position error

$$\Delta r = \sqrt{(x - x^d)^2 + (y - y^d)^2}$$

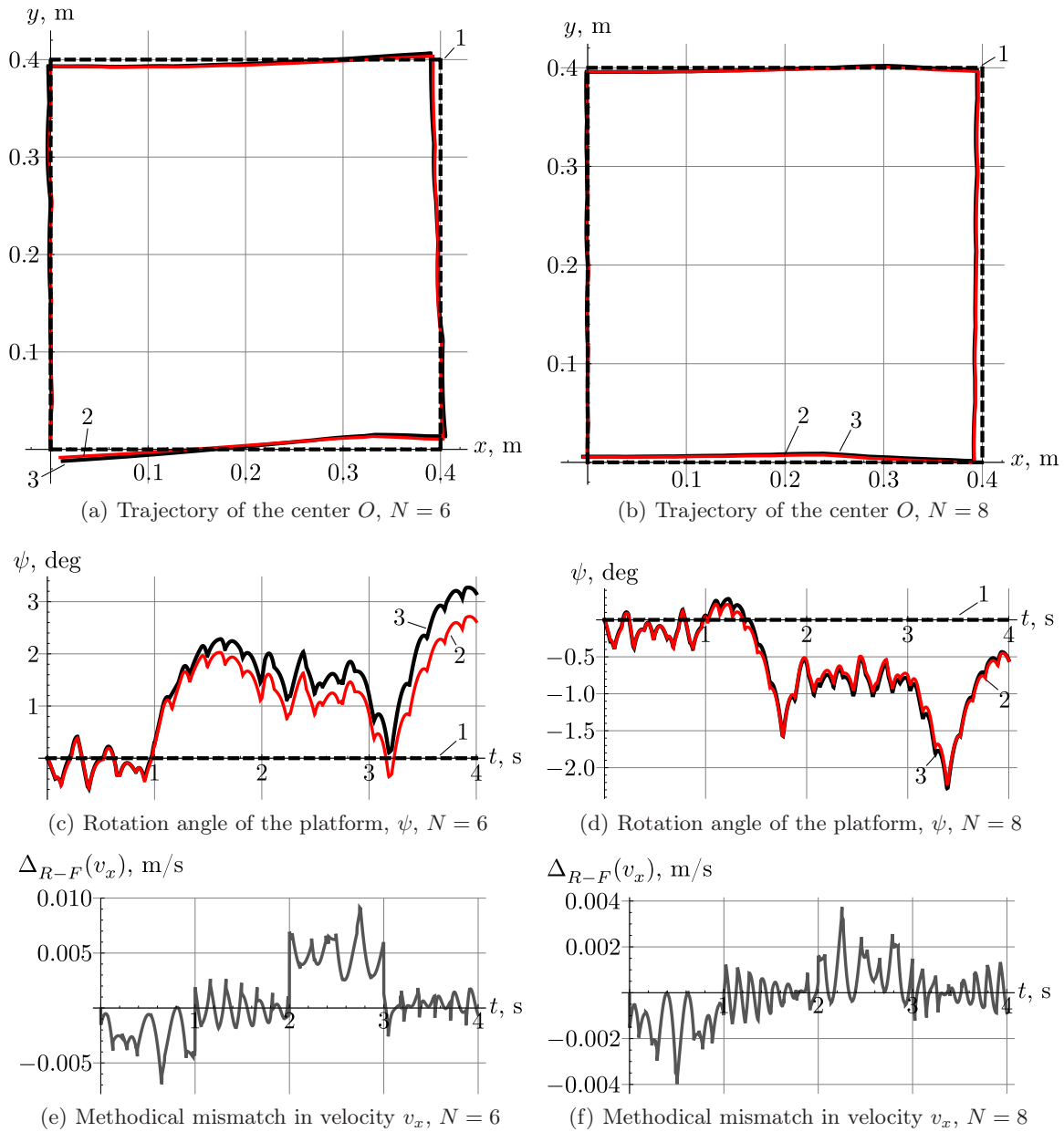


Fig. 14. Motion along the square: comparison of the kinematic equations (parameters (II)). 1 — desired motion (A); numerical simulation results using: 2 — the “full-size” Eqs. $\omega_{Fi} = \dot{\varphi}_i^d$; 3 — the “reduced” Eqs. $\omega_{Ri} = \dot{\varphi}_i^d$

and its maximum value on the aggregate

$$\Delta r_{\max} = \max_{0 \leq t \leq t_f} \max_{\Phi_0} \Delta r;$$

- an angular error

$$\Delta \psi = \psi - \psi^d$$

and its maximum value on the aggregate

$$\Delta\psi_{\max} = \max_{0 \leq t \leq t_f} \max_{\Phi_0} |\Delta\psi|;$$

- average errors along the trajectory length

$$\Delta r_{\text{av}}^{\ell} = \frac{\Delta r_{\max}}{\ell}, \quad \Delta\psi_{\text{av}}^{\ell} = \frac{\Delta\psi_{\max}}{\ell},$$

where ℓ is the length of the reference trajectory ($\ell = 8\rho$ for motion (A), $\ell = 2\pi\rho$ for motions (B)–(D));

- estimates of a maximum error growth rate along the trajectory length

$$\Delta r_{\max}^{\ell} = \frac{\Delta r_{\max}}{\ell_{\Delta r_{\max}}}, \quad \Delta\psi_{\max}^{\ell} = \frac{\Delta\psi_{\max}}{\ell_{\Delta\psi_{\max}}},$$

where $\ell_{\Delta r_{\max}}, \ell_{\Delta\psi_{\max}}$ are, respectively, the path traveled along the reference trajectory when reaching the maximum Δr and $|\Delta\psi|$.

Note that in all the cases under consideration, the maximum of $\Delta\psi$ on the aggregate of the initial conditions Φ_0 is reached at $t = t_f$ or $t \approx t_f$, which is why

$$\Delta\psi_{\max}^{\ell} \approx \Delta\psi_{\text{av}}^{\ell}.$$

Table 1. Accuracy of the desired motions tracking

Desired motion	Geometric parameters	N	$\Delta r_{\max},$ mm	$\Delta r_{\text{av}}^{\ell},$ mm/m	$\Delta r_{\max}^{\ell},$ mm/m	$\Delta\psi_{\max},$ deg	$\Delta\psi_{\text{av}}^{\ell},$ deg/m
(A)	(I)	6	17.0	10.6	21.1	1.7	1.1
		8	9.0	6.1	11.9	1.0	0.6
	(II)	6	35.0	21.9	21.8	5.1	3.2
		8	13.6	8.5	10.8	2.3	1.4
(B)	(I)	6	19.4	15.4	24.3	1.86	1.5
		8	11.8	9.4	14.7	1.42	1.1
	(II)	6	30.0	23.8	23.9	7.0	5.6
		8	18.2	14.5	14.6	3.9	3.1
(C)	(I)	6	14.8	11.8	22.9	0.85	0.7
		8	8.5	6.8	12.6	0.43	0.3
	(II)	6	18.5	14.7	25.7	3.9	3.1
		8	10.2	8.1	15.8	2.2	1.8
(D)	(I)	6	97.6	77.7	110.2	23.1	18.4
		8	69.2	55.1	76.2	17.4	13.8
	(II)	6	76.9	61.2	112.8	9.3	7.4
		8	36.8	29.3	68.5	5.1	4.1

The maximum desired motion tracking errors, as well as estimates of their growth rates, are given in Table 1.

The robot's motion is illustrated by graphs of the trajectories of the center O , the angles of the platform rotation ψ , and the positional error Δr (Figs. 15–18). The areas in which the entire set of graphs is located for $\varphi_i(0) \in \Phi_0$ are shown, as well as two examples of curves corresponding to specific values of the initial wheel rotation angles.

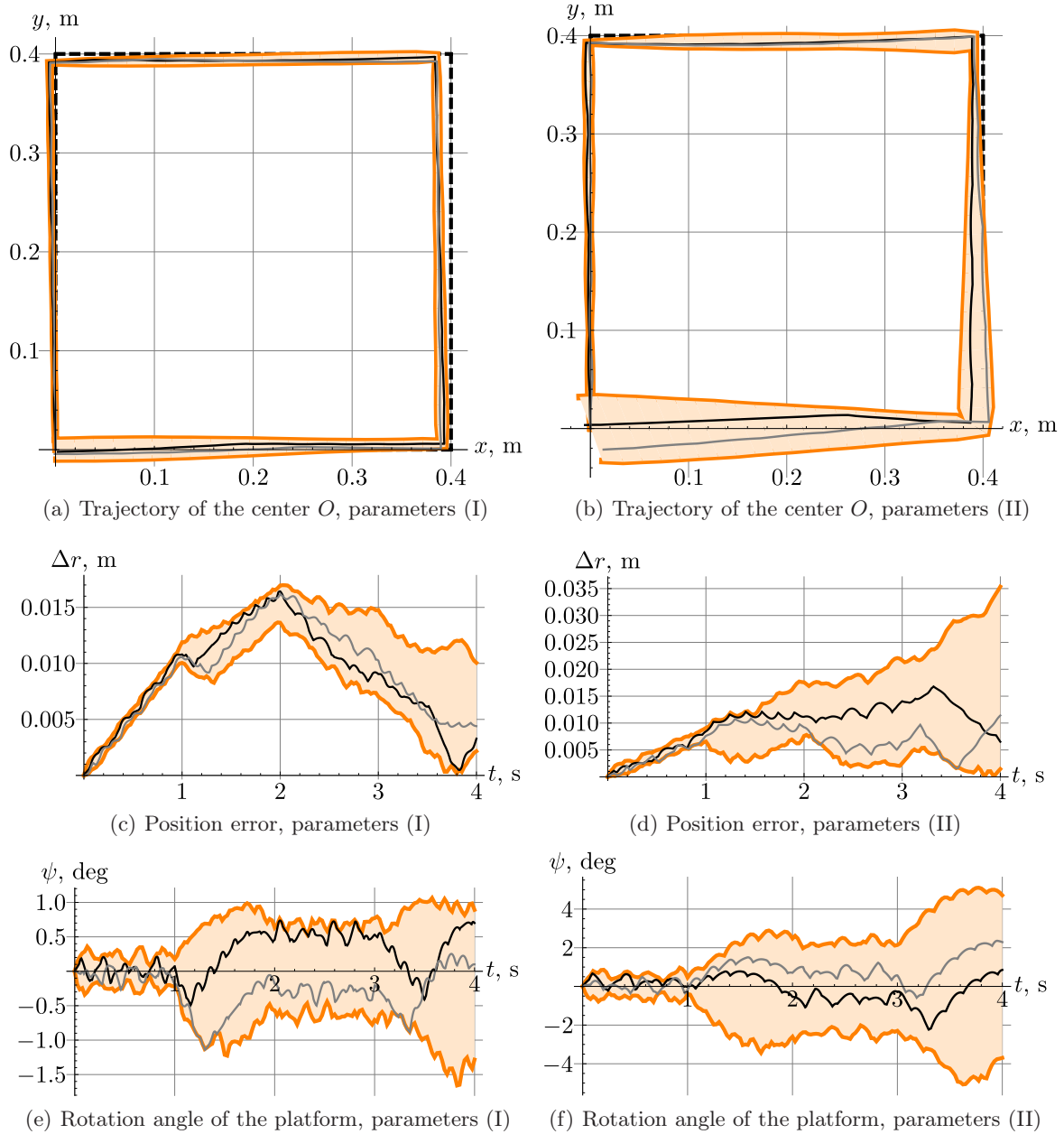
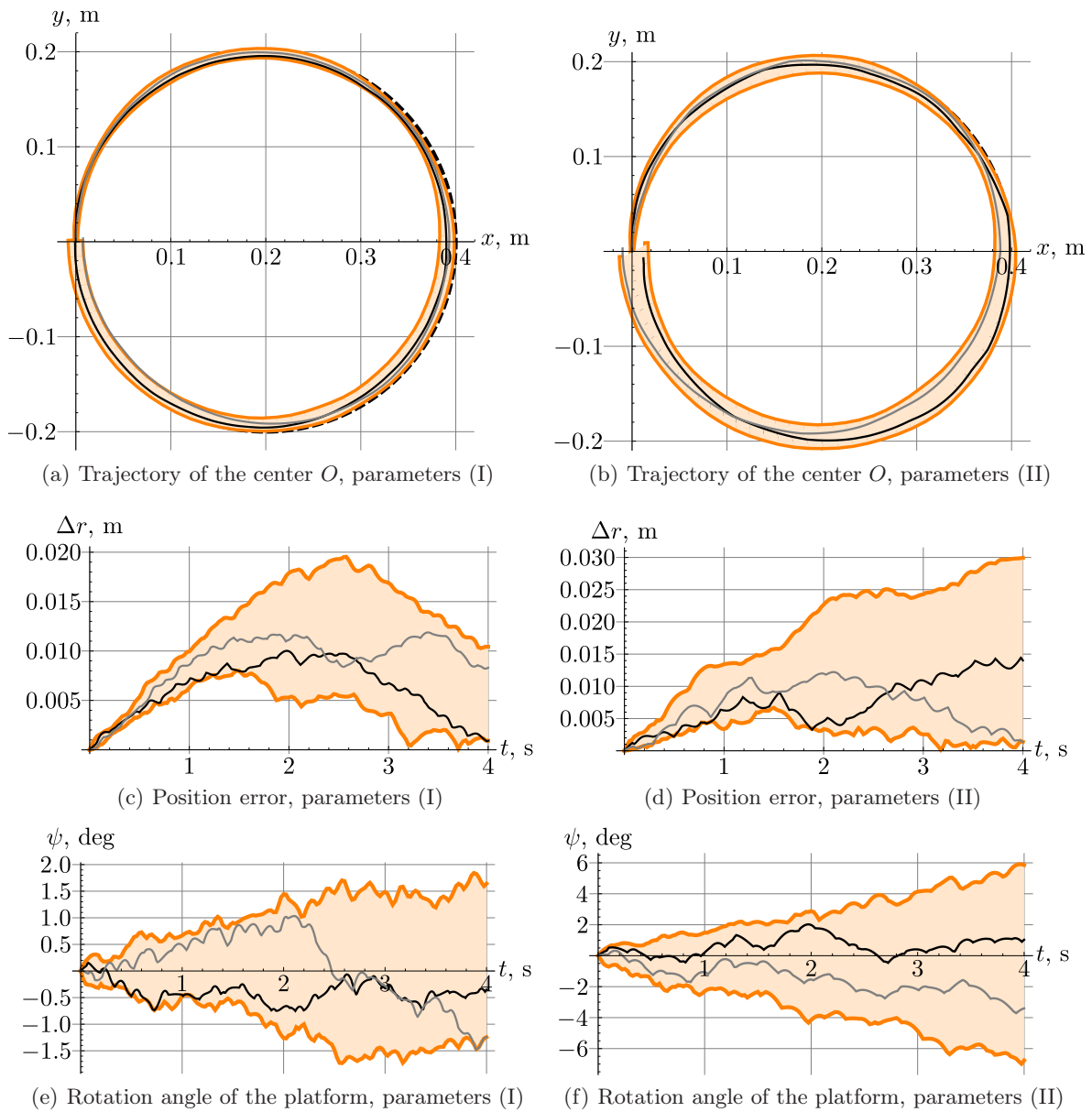


Fig. 15. Accuracy of the motion (A), $N = 6$

The simulation results show that increasing parameter κ and decreasing N (that is, increasing the magnitude of $\tilde{\varphi}_i$) leads to a decrease in the accuracy of the motion.


 Fig. 16. Accuracy of tracking of motion (B), $N = 6$

When tracking motions (A), (B), (C), positional errors with the same combinations of the geometric parameters and the number of rollers do not differ much from each other. Here the rate of growth of the positional error is $\Delta r_{\max}^{\ell} \approx 21 \dots 25$ mm/m for $N = 6$ and $\Delta r_{\max}^{\ell} \approx 11 \dots 15$ mm/m for $N = 8$ ($\Delta r_{\max}^{\ell}$ depends weakly on other parameters, see Table 1).

However, for motion (C) (the longitudinal motion along the circle, Fig. 17), the main deviation Δr is achieved in the direction of the trajectory, and for motions (A) and (B), in the lateral direction relative to the trajectory (Figs. 15, 16).

REMARK 5. The qualitative appearance of the graphs $\Delta r(t)$, $\Delta \psi(t)$ for motions (A)–(D) is approximately the same; for motions (C), (D) these graphs are not given.

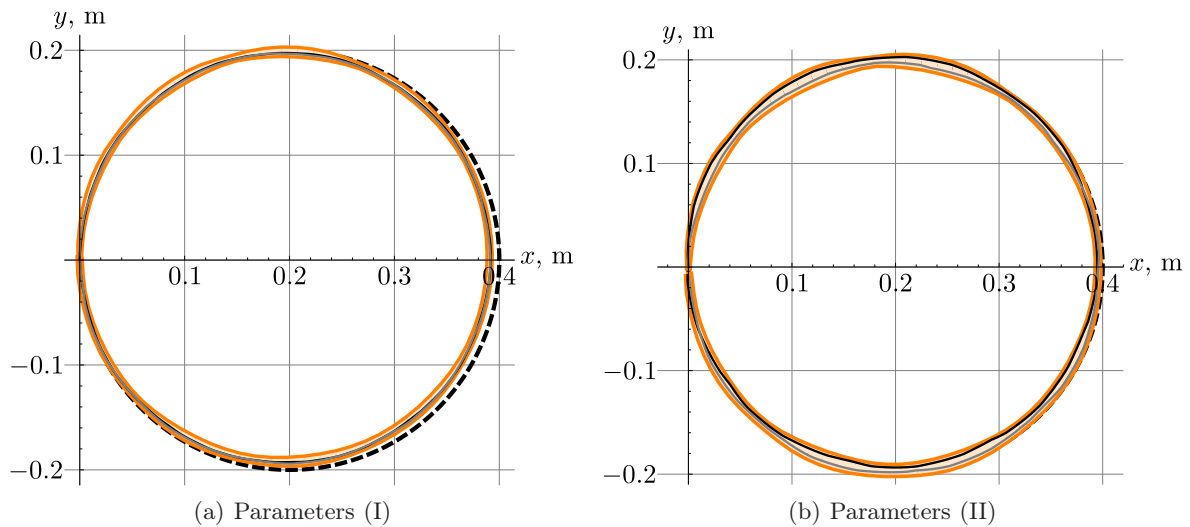


Fig. 17. Trajectories of the robot center when tracking the longitudinal motion along the circle (motion (C), $N = 6$)

The positional errors and their growth rates when tracking motion (D) (the transversal motion around a circle, Fig. 18) are four times greater than those for other motions. Errors in the platform rotation angle $\Delta\psi$ also increase many times over. For geometric parameters (I) and $N = 6$, the maximum error in angle for motions (A), (B), (C) is $\Delta\psi_{\max} \approx 1^\circ \dots 2^\circ$, but for motion (D) it is $\Delta\psi_{\max} \approx 23^\circ$.

Thus, the accuracy of the desired motion tracking depends not only on the geometric parameters of the robot, but also on the combination of velocity and direction of motion, and the speed of rotation of the platform. In some cases (for example, for motion (D)), a “resonance” occurs, in which the rate of the error growth becomes significantly greater than that for the “nonresonant” motions (for example, motions (A)–(C)).

It is also possible to distinguish the “antiresonance” motion (C), during which the lateral deviation from the reference trajectory is significantly less than that in other cases.

7. Conclusion

In this article, the kinematic relations for the Mecanum wheels moving on a flat or a spherical surface are obtained. From the condition of continuous contact with the supporting surface $V_{Pn} = 0$, the equations for the displacement of the contact point in the direction of the wheel axis were obtained, which were used to study the geometry of the wheel rollers.

In contrast to studies [19, 20, 27, 28], the geometric relationships for the roller were obtained based on the analysis of the kinematics of velocities, which made it possible to quite easily generalize the technique to the cases of the wheel motion on a sphere or in it.

The kinematic equations obtained in Section 3 for the motion of the wheel on the internal surface of a sphere can be considered, in a sense, universal:

- the substitution

$$\kappa \rightarrow 0, \quad \lambda \rightarrow 1, \quad R_T \rightarrow R_S \rightarrow \infty$$

gives similar equations for motion on a plane;

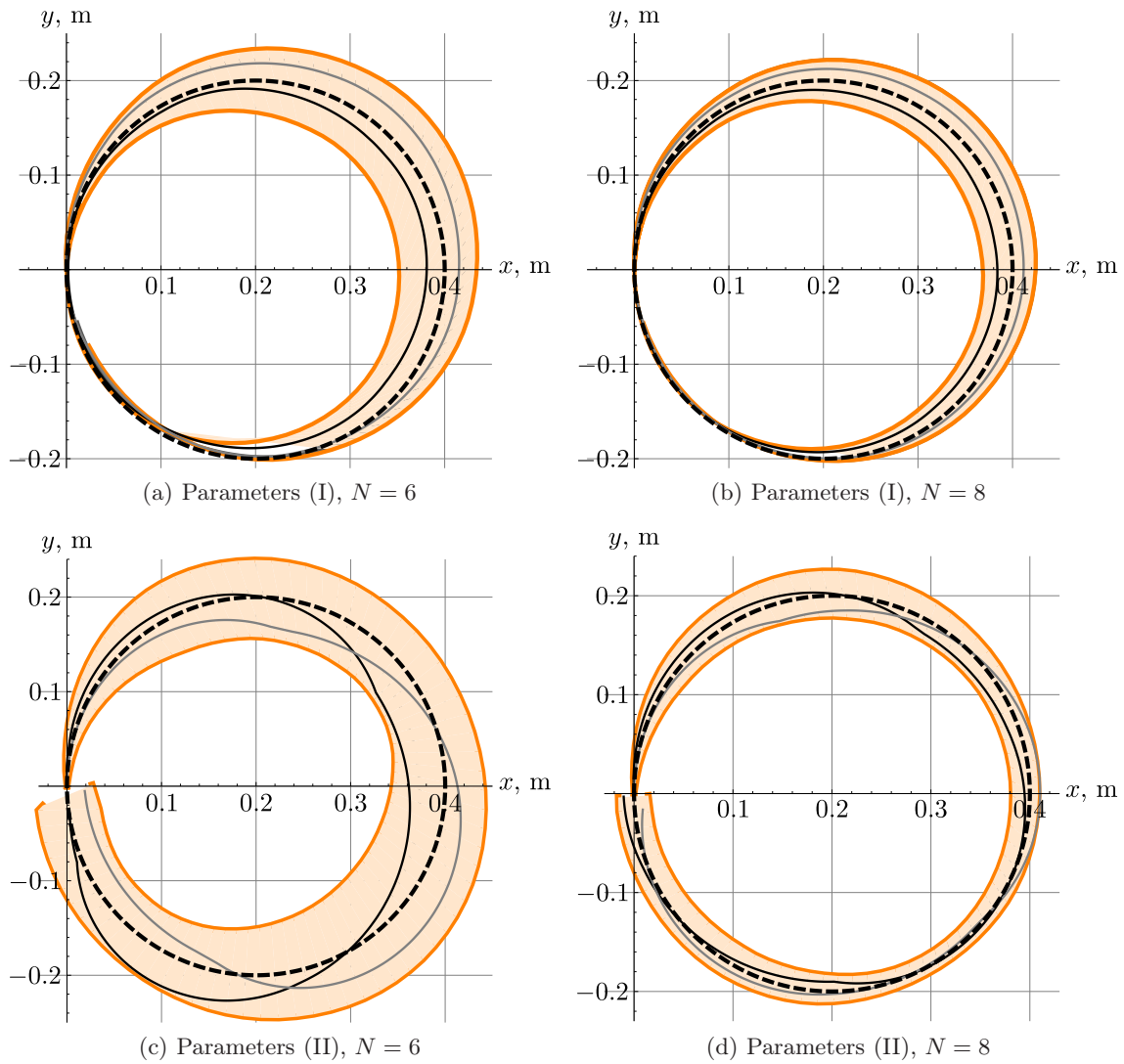


Fig. 18. Trajectories of the robot center when tracking the lateral motion along the circle (motion (D))

- the substitution

$$\kappa \rightarrow -\kappa, \quad R_S \rightarrow -R_S, \quad R_S - R = R_T \rightarrow -R_T = -R_S - R$$

gives similar equations for motion on the external surface of a sphere.

The equations for the rotational velocities of the wheels and the rollers $\dot{\varphi}$, $\dot{\gamma}$ are necessary for analyzing the motion of systems within the framework of nonholonomic models, solving problems of controlling Mecanum systems and increasing their accuracy.

The equations for the velocity components V_{P_X} , V_{P_T} of the contact point and the angular velocities of the roller relative to the supporting surface Ω_{r_X} , Ω_{r_T} , Ω_{r_n} are necessary to describe the dynamics of Mecanum systems taking into account the contact friction forces and moments in the presence of slipping.

The geometric relationships for the rollers of the ideal Mecanum wheels will help to adjust the existing shape of the Mecanum wheel rollers of spherical robots and ballbots to improve the

contact conditions between the rollers and the spherical surface. The need for such a correction is evidenced by the results of practical studies [9, 28].

The results of the numerical simulation of the motion of the spherical robot with the internal three-wheeled Mecanum platform show a decrease in control accuracy in the absence of feedback on the robot position due to effects associated with a finite number of the rollers, their geometry and switching. These effects lead not only to high-frequency vibrations, but also to a “drift” of the robot trajectory relative to the desired one. Compensation of this effect by means of positional feedbacks requires an additional energy consumption.

The qualitative appearance of the trajectories of the spherical robot is consistent with the experimental results [6–8].

The simulation results show that the accuracy of the desired motion tracking is affected not only by the geometric parameters of the rollers, but also by the ratio of the velocity components of the Mecanum platform, which in some cases leads to a significant increase in errors.

Further research on this topic involves the use of motion separation methods and statistical methods for analyzing the kinematics and dynamics of Mecanum systems.

Appendix A. Comparison of the geometry of the Mecanum wheel rollers on a sphere

Let us compare Eqs. (5.1)–(5.4) describing the geometry of the rollers with similar results obtained in [28]:

$$\begin{aligned}\zeta^{\text{pln}} &= \zeta^{\text{ext}} = \left[S + \frac{1}{2} \left(\left(\sqrt{R^2 - S^2} - R_H \right) \frac{-S}{\sqrt{R^2 - S^2}} \right) \right] \sqrt{2}, \\ \xi^{\text{pln}} &= \sqrt{\left(\sqrt{R^2 - S^2} - R_H \right)^2 + \frac{1}{2} \left(\left(\sqrt{R^2 - S^2} - R_H \right) \frac{-S}{\sqrt{R^2 - S^2}} \right)^2}, \\ \xi^{\text{ext}} &= \xi^{\text{pln}} + R_S - \sqrt{R_S^2 - \left(S + \frac{1}{2} \left(\left(\sqrt{R^2 - S^2} - R_H \right) \frac{-S}{\sqrt{R^2 - S^2}} \right) \right)^2},\end{aligned}$$

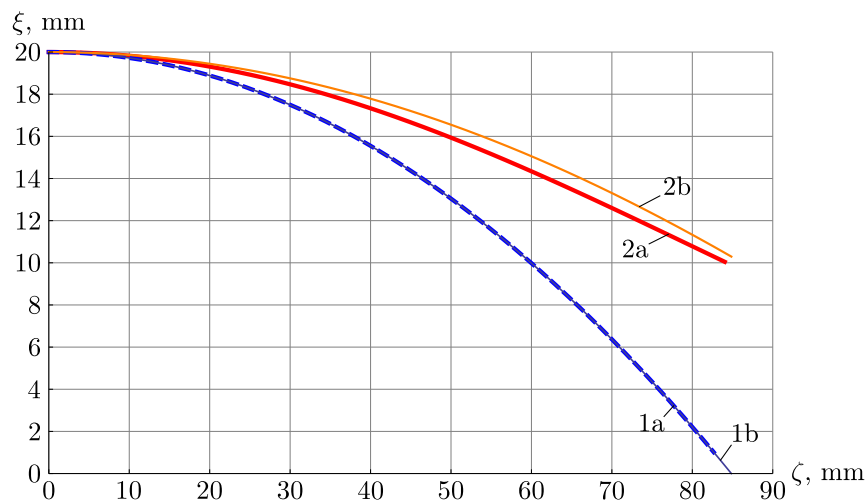


Fig. 19. Roller generatrices for the Mecanum wheel ($R = 100$ mm, $R_H = 80$ mm) on a plane or on a sphere external surface ($R_S = 180$ mm). Wheel on a plane: 1a — Eqs. (5.1), (5.2); 1b — Eqs. [28]; wheel outside a sphere: 2a — Eqs. (5.3), (5.4); Eqs. [28]

where S is a parameter. Here the formulas [28] for ζ^{ext} and ξ^{ext} are approximate (in particular, due to the assumption of the authors about the coincidence of $\zeta^{\text{pln}} = \zeta^{\text{ext}}$, see the review in the introduction).

Let us consider the numerical example from [28] (Fig. 19). The generatrices of the rollers for moving on the plane, constructed according to formulas (5.1), (5.2) and the equations from [28], coincide. But similar results for the Mecanum wheel on a sphere differ (by less than 1 mm, i. e., less than 5 % of the central cross-section radius).

Conflict of interest

The author declares that he has no conflict of interest.

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